

Opportunity Lifecycle Intelligence

Current-truth reconstruction,
scientific design, and execution master plan

What is developing?
What can be considered now?
What remains from this landmark?
What does the existing plan need next?

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Prepared for Chairman Chris

Recommended architecture and governed execution plan.

No statistical promotion, production certification, or trading authority granted.

Contents

The central decision: recover and connect the existing owners. OLI belongs in Alpha Intelligence's governed opportunity semantics and Market OS's product composition, with TOI, Prophet, Radar and the native evidence owners retaining their truth.

Evidence key: I = internal source or receipt; X = research paper; P = official product documentation; C = Chairman commission. Direct observations, reported historical receipts and recommendations are distinguished throughout.

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Reading guide and evidence standard

The commission requires a recovered program, not a fresh program name. This report follows its 22 deliverables, with an explicit source register and a separate executable handoff. The source census preceded external research. Internal references use **I** identifiers, papers use **X**, vendor documentation uses **P**, and the uploaded commission is **C00**. The register records source location, access level, dates, and limitations. [C00]

Direct observation means a current connector response or code fragment was read. **Reported receipt** means a current source describes an earlier test, experiment, deployment, or acceptance; this run did not reproduce it. **Proposal** means this report's design judgment. **Unknown** means the required evidence was not established. These distinctions apply even when a historical receipt lives on current main.

This was a targeted primary-source research review, not an exhaustive systematic review or a pooled meta-analysis. The literature spans different instruments, frequencies, objectives, and eras; its effect sizes are not transferable estimates for Mastermind. Some papers were available as full text, others only through publisher or author abstracts. The source register makes that distinction. Product claims were checked against official documentation, not through authenticated hands-on testing. No market-outcome backtest, protected outcome read, live trade, source change, project mutation, worker dispatch, or production browser recertification was performed in this commission.

1. Executive conclusion

1.1 Recommended definition

Opportunity Lifecycle Intelligence (OLI) should be the governed, current-landmark interpretation of an opportunity across its native evidence, episode, strategy, and user-state owners. It should explain phase and change, estimate transition-specific outcomes where qualified, distinguish present entry from holding, and preserve every accountable origin and subsequent decision.

Choose **architecture B, refined through already-existing owners**. Do not generalize TOI into the sovereign owner of all opportunity truth. Do not introduce an OLI database, scanner, event identity, score, species registry, or independent orchestration lifecycle. The census found that Alpha Intelligence already owns cross-domain opportunity semantics, an Opportunity Evidence Vector already has executable composition semantics, and Market OS already owns the compact security-state product projection. The proposed OLI contract belongs within those boundaries. [I16, I17, I18, I19, I32]

Canonical semantic home: WS: ALPHA- INTELLIGENCE- INTEGRATION, through its existing K3 opportunity semantics and any separately admitted K5 OpportunityCase extension. **Technical occurrence and technical lifecycle science:** WS: TECHNICAL- OPPORTUNITY- INTELLIGENCE. **Product composition:** WS: MARKET- OS, with Prophet and Terminal as governed consumers. **Prophet identity, candidate-state projection, and current entry permission:** the incumbent B1, B3, and B4 owners. This is an architectural recommendation grounded in the existing ownership decisions, not a record that those owners have already adopted a new OLI schema. [I18, I20, I22]

1.2 What changes the original thesis

The original brief correctly emphasizes transitions and remaining opportunity, but understates the existing cross-owner architecture. OLI is not missing because nobody created an appropriately named workstream. It is missing as an **integrated, qualified decision experience** across already-existing semantics, partially accepted research, incomplete transport, and product projections.

Three further corrections matter. First, pre-trigger observations cannot be forced into Prophet's canonical episode registry merely to obtain an identifier. Current B1 code admits only `turn_watch_reset_low`, enforces one active episode per security/identity epoch, and rejects unknown anchor kinds. Second, TOI W0 does not authorize a new durable occurrence-history database. Third, K5 OpportunityCase dependencies cannot be bypassed by calling a new composer "OLI." [I04, I16, I18, I22]

1.3 The first product and first science

The first useful product is an **honest opportunity context view**: native phase, lawful entry status, current quote and geometry basis, known change, one important support, one important contradiction, missing data, and next observable condition. It can be useful with unsupported forecast heads explicitly null. It must never manufacture permission or probability to make an attractive interface.

The first science is a **same-information, full-population landmark comparison**: does updating an assessment at a registered decision time improve conditional forecasts and policy value over the current static classifier, TOI's two queues, and existing Prophet behavior? Test remaining-entry opportunity separately from later holding fragility. More expressive models come after transparent hazards and fixed-policy controls, not before.

1.4 Immediate decision

Recover #7107 and #7094, reconcile stale organizational projections, and freeze a small cross-owner contract with an explicit identity join law. In parallel, prepare fixtures and a read-only product slice using admitted facts. Keep all technical outcome research behind its data/scientific gates, preserve the live Fable owner on #784, and preserve Top Anatomy's natural evidence wait. A report can finish the architecture commission without pretending those empirical gates have finished. [I05, I06, I07, I08, I09, I10, I11, I12, I13, I14, I15, I23, I24, I25, I26]

2. Current-truth census

2.1 Source and runtime identities

Surface	Observed identity	Meaning and limitation
Protected Mastermind master	50a7a5d31854c1161335e2b6ffdb04eb268fe6d	Current source reference read for this census; supersedes the brief's navigation pin for procedure reads.
Macro main	6f5e78e94e8808582a650cdfa0fc3357040a179c	Matches the brief's Macro snapshot. Source existence does not imply live serving or data admission.
Terminal master	41b8af2da46614cedd2a485214e53003d4f030fc	Current remote reference observed, separate from any local worktree head.
Executive installed Mastermind	b3627c580dd37ac1c167f59ac4b7555a4330edef	Runtime observation at 08:56:52 UTC on October 3. Not the same identity as current protected source.
Executive installed Macro	88804ed7079700c598bb8e04aa64307d1335402d	Installed runtime source, not a claim about the website's served release.

The reads were sequential, not a cross-repository atomic snapshot. A hash mismatch alone does not establish ancestry, regression, or deployment failure. The Executive response had no running jobs in its returned view, but this does not contradict separately evidenced external Fable execution. At any modifying or release boundary, refresh the exact affected source, writer custody, and runtime binding rather than treating these pins as perpetual authority. [I01, I02, I15, I31]

2.2 Technical Opportunity Intelligence

W0 exists and is merged. TOI's architecture already distinguishes anticipation from triggered actionability, identifies trigger/invalidation and confirmation/chase costs, requires contradictions and remaining-opportunity treatment, preserves technical primitives and Setup Species, and gives Radar tactical ownership. Current Agent OS correctly records W0 done through #6570 but incorrectly leaves W1 and W2-0 undispached. [I03, I04]

W1 exists but is not accepted. #7107 is open, draft, and held at head 15e4dfb0c9ddc788127d0954fd7d92414eefdcbf. Its 32 method passports, 29 equivalence classes, ten residual-family dispositions, reproduction material, and independent review are substantial reusable work. Four proposed W3 families and 32 constructions are not 32 accepted hypotheses; the draft material has zero new W3 registrations. Preserve prior closed source-quality findings rather than commissioning a new census. [I05, I06]

The committed coverage partition and the proposed repaired partition are different facts. Existing committed reporting is 26 exact / one partial / five missing. The bounded repair proposes 24 exact / three partial / five missing after correcting Bollinger and Donchian bindings. The latter is not yet committed acceptance. Current repair work must preserve the incumbent uncommitted AST changes, isolate the intended duplicate-membership guard, bind bandwidth to its real numeric primitive, and retain the frozen-occurrence/latest-break mismatch as an honest partial. [I05, I06]

W2-0 exists but the broad substrate is not admitted. #7094 is open/draft at 5bb1bc68c99146fab040aade04bbf1903c51e5b7. Its reviews distinguish semantic acceptance of a records-only archaeology result from release and data qualification. Reported measurement found 15/15 regular-session 4H parity passes and 5/5 early-close failures, including November 28, 2025. Provider-built hourly fallback starting at 10:00 ET cannot establish exact 09:30-open whole-universe parity. Daily/intraday adjustment compatibility and historical eligibility remain unresolved. A future merge accepts a PARTIAL/HOLD record, not permission to run W3. [I07, I08]

Elliott E0 remains specification only. #7069 is open/draft at 3aa54c741aa6218379ca9498069219f84d876d2b, explicitly held for W1/W2-0. Reuse causal pivot timing, ambiguity/abstention, and equal-budget generic geometry controls. Do not call this a validated wave-recognition or timing system. [I09]

2.3 Tactical events and Dislocation/Reclaim

Live Entry Radar remains the incumbent 5-minute event and evaluation owner. MAS-25 is In Review: W6 source implementation exists, while final bounded post-merge acceptance remains owed in the retrieved portfolio record. Source implementation, private transport, forward grading connectivity, and product acceptance are separate capabilities. [I12]

Terminal #598 is the Tactical Intelligence parent. #784 has actual Fable **pickup and START receipts**, not merely a proposed assignment. The live incumbent is auditing and implementing Dislocation/Reclaim. Its audit reports a broken intake seam: W4 writes to R2 while W5's reconciliation path expects a local directory absent on its runner. Therefore a waiting status cannot establish that the market produced no events. This is an active-owner finding requiring that owner's repair and proof; this research did not reproduce the runtime defect. [I13, I14, I15]

The same audit found existing R1 carriers, registered constructions, and unpushed work. It also identified catalyst freshness/schema problems in #8305 and a held private-evidence transport boundary. OLI must consume the eventual versioned facts and receipts, not create another reclaim detector, catalyst repair, consumer, spool, or worker session. [I15]

2.4 Prophet, identity, and management

R6 already separates member selection, entry timing, management, origin forecasts, current-landmark forecasts, source clocks, and strategy-native hold laws. These are not inventions of this report. R6 itself describes a proposed integrated plan and governed execution, not blanket delivery or statistical acceptance. [I20]

The current B1 core is stronger than its September 23 census: its anchor vocabulary is now closed. It admits only `turn_watch_reset_low`, rejects unregistered anchors, prevents a different anchor from silently replacing an active episode, preserves ordinary source-key immutability, and ties opening time to when the anchor is known. A missing structural anchor cannot be repaired with a ticker/date surrogate. Radar events attach by exact native identity; the Radar runtime episode and canonical Prophet episode are not aliases. [I21, I22]

Historical R6 census material also identifies several incompatible uses of “episode,” including options-flow episodes and plan-book ticker ordinals. That diagnosis remains a useful integration warning. Its historical statement about the grader's board-level rather than B1-episode coupling was not re-executed against current production, so the exact current grading join is an explicit acceptance check, not an asserted newly discovered live defect. [I21]

Phase 22 already froze a distinction between consumed opportunity at entry and hidden fragility during the hold. It requires a real forward transport epoch, same-cut C4 context, source-qualified real-yield knowledge clocks, and TrialLedger registration before protected outcomes. Its existing floors—30 episodes, 12 decision months, and name effective-N of at least eight per arm—must not be lowered, borrowed as universal OLI sufficiency, or retrospectively declared satisfied. [I23]

2.5 Cross-domain and product infrastructure

Component	What exists	What OLI may consume now	What is not established
Alpha K3-E / OEV	Accepted contract/composer carrier #6417; typed provenance, nulls, gates, authority separation	Admitted owner facts through the existing deterministic view contract	A new lifecycle extension, all live consumers, or K5 acceptance
Market OS	Security-state product owner; historical AAPL B1A live acceptance; existing Portfolio/Watchlist owners	Compatible admitted projections and source-backed historical acceptance receipts	General-universe expansion and fresh October 3 served-path conformance
GMI/theme	Graph substrate, memberships, research and active theme verticals	Qualified source-native membership/breadth/context with its own clocks and eligibility	Universal accepted ThemeState or complete historical membership vintages
Earnings/D5	Narrow owner-backed event adapter and reported AAPL dossier proof	Existing allowlisted facts and typed absence, through revision-aware readers	Whole-universe event/PIT coverage, unlicensed consensus, automatic beat/miss inference

Component	What exists	What OLI may consume now	What is not established
Options	Measured trade/NBBO evidence; canonical episode/campaign owners; Oct 2 retrospective research package	Qualified descriptive witness; separately labeled retrospective associations	Directional intent, dealer inventory, OOS/PIT promotion, complete publication acceptance
Evaluation / Species	Existing trial, claim, outcome, species-promotion and arena owners	Native contracts and legally admitted evidence	Any new OLI forecast's calibration or authority

These dispositions come from current source reads of contracts and workstreams, including historical receipts explicitly dated inside them. They are not a fresh universal runtime-health certificate. [I16, I17, I18, I19, I20, I27, I28, I29, I30, I32]

2.6 Top Anatomy and Temporal Grain

Top Anatomy's R0 restoration is recorded complete. Its remaining confirmatory out-of-time question is intentionally waiting for natural maturity; the workstream places earliest eligibility roughly in July 2027. Do not reopen its ruler, relabel an immature null as failure, or commission repeated outcome peeks. Maturation and deterioration descriptors may be consumed only at their existing evidence level. [I24]

Temporal Grain is a sibling clock research program. Its W0 record still says awaiting CI, while GitHub #6790 is merged. The merge proves a specification release, not useful adaptive clocks. W1A mechanical survival, W1B usefulness, later outcome-blind scale derivation, and instrument-disjoint confirmation remain distinct. It does not replace TOI W2's broad data/session admission. [I25, I26]

3. Orphan and staleness map

3.1 How fragmentation occurred

The evidence supports a **coordination and acceptance failure**, not a finding that the organization never researched opportunity lifecycles. Several programs solved different parts of the same journey under legitimate owners. Their records then diverged: mutable PR prose retained old instructions, accepted source advanced without every organizational projection following, runtime transport remained incomplete, and related research accumulated without a single customer-facing decision contract.

This is a diagnosis from the observed discrepancies, not a claim to know every historical organizational cause. The corrective response is to reconcile receipts and interfaces, not centralize all systems under a new name.

Discrepancy	Adjudication	Recovery action
P-MAS-54 has zero children; #7107/#7094 exist	Stale/incomplete portfolio projection	Project the existing carriers, preserving their real holds; do not create replacement research.
TOI Agent OS says W1/W2 undispached	Stale main record	Reconcile exact carrier/head/status and acceptance debt; #7094's unmerged correction explains part of the divergence.
#7094 has semantic PASS but remains draft	Not contradictory	Records semantics, release checks, data qualification, and W3 permission are distinct.

Discrepancy	Adjudication	Recovery action
#6417 body contains HOLD while PR is merged	Stale mutable description	Use merge metadata and accepted contract scope; do not infer consumer activation.
Temporal W0 says awaiting CI while #6790 is merged	Stale projection	Correct W0 release state; retain W1A/W1B gates.
#784 old handoff has no receiver, later comments have START	Superseded handoff	Preserve the actual active Fable incumbent and its worktree/carriers.
B02 census describes open anchor vocabulary; current code closes it	Repaired historical defect	Do not redo the allowlist repair. New anchors still need owner admission.
P-MAS-51 has zero children, Top R0 complete	Incomplete projection, not abandoned science	Record restoration plus deliberate maturity wait; no new OOT experiment.
Radar waiting status with no working intake path	Operational non-observation	Establish transport health before interpreting absence as “nothing fired.”

Sources: [I03, I05, I06, I07, I08, I09, I10, I11, I12, I13, I14, I15, I16, I17, I21, I22, I24, I25, I26].

3.2 Program classification

TOI is **reuse + dependency + stale projection**. W1/W2/E0 are **active incumbent carriers**, not missing work. Prophet's origin/current-landmark and plan distinctions are **absorbed semantic concepts with retained ownership**. Radar is **dependency + active incumbent**. #598 is a **product parent** and #784 an **active incumbent**. Top Anatomy and Temporal Grain are **siblings**, not takeover targets. OEV and Market OS are **reuse**, and a new OLI composer/store would be **duplicate risk**. The unqualified landmark history join, permitted pre-trigger denominator persistence, broad B1 anchor admission, transition forecast evidence, and unified production experience are **genuinely missing or unproven capabilities**, depending on the exact downstream carrier. [I03, I04, I05, I06, I07, I08, I09, I10, I11, I12, I13, I14, I15, I16, I17, I18, I19, I20, I21, I22, I23, I24, I25, I26, I27, I28, I29, I30, I31, I32]

No entire related program should be marked obsolete from this census. Obsolete items are specific projections or superseded assumptions, such as the older open-anchor defect description. A stale row is not authority to cancel a live worker.

4. Ownership architecture and no-duplicate map

4.1 Adjudication of A, B, and C

Criterion	A: broaden TOI into OLI owner	B: incumbent semantic federation	C: independent OLI engine/store
Ownership	Collides with Alpha semantics, Prophet decisions and Market OS product	Preserves one writer per truth; requires explicit contract adjudication	Creates a new competing sovereign owner

Criterion	A: broaden TOI into OLI owner	B: incumbent semantic federation	C: independent OLI engine/store
Identity	Risks treating technical occurrences as all opportunities	Keeps native IDs and admits typed relationships	Tempts a universal replacement episode ID
Recompute/history	May absorb other owners' clocks	Derived read model; append predictions through existing ledgers	New event/history/replay machinery
Scientific evaluation	Useful for technical transitions, insufficient for all sleeves	Transition studies use existing Species/Trial/Evaluation	Parallel trial/promotion risk
Migration	Broad semantic replacement	Versioned adapters and bounded projections	Large migration and backfill temptation
Operational burden	Ownership concentration and collision	More interface discipline, less duplicate infrastructure	Highest custody, synchronization and rollback burden
Decision	Reject as the cross-domain architecture	Recommend, with explicit existing-owner home	Reject absent a demonstrated unfillable gap

This is a qualitative architectural judgment, not a measured cost benchmark. B's real cost is interface discipline: incompatible identities and knowledge clocks cannot be hidden behind a convenient join. That cost is necessary correctness, not a reason to choose a new database. [I04, I16, I18, I20, I22]

4.2 Exact owner responsibilities

Canonical truth	Incumbent owner	OLI boundary
Technical numeric primitives	tech_catalog and native primitive implementations	Reference primitive/version/units; no duplicate indicator implementation.
Technical occurrence and transition definition	TOI + registered technical species	Consume native state; propose new mappings only through that owner.
Scientific species and validation/deployment lifecycle	Setup Species / species_registry	Read status and evidence; no shadow registry.
Trial families, search budget, outcomes and promotion	TrialLedger / QLedger / Evaluation OS / existing arena	Extend native record schemas only for demonstrated missing relations.
Security, issuer, listing, price basis and transport	Data OS and market-data owners	No ticker surrogate or new market-data plane.
Canonical Prophet episode identity/re-arm	B1 candidate episode owner	Exact episode/generation reference or explicit absence.
Prophet candidate-state projection	B3	No competing global phase scorer.

Canonical truth	Incumbent owner	OLI boundary
Present strategy-specific entry permission	B4 and accepted strategy definitions	Pass through owner verdict/reasons; forecasts never waive gates.
Plan and actual position state	Existing Plan, WatchStore, Portfolio/Risk owners	User-specific join stays private; a save is not a fill.
Tactical 5-minute events and grading	Live Entry Radar	No second scanner, spool, WebSocket, evaluator or forward event ID.
Cross-domain opportunity semantics	Alpha Intelligence K1/K3/K5	OLI is a governed profile/extension here, not a new workstream.
Compact security-state product	Market OS	Existing composer/transport/cache; Terminal renders rather than mints truth.
Theme/member/breadth	GMI and native group owners	Preserve PIT membership, denominator and rights.
Catalyst, earnings, company evidence	Native event/company owners	Preserve source revision and expectation comparability.
Options evidence and campaign identity	ThetaData/live-flow/options owners	Incremental witness, not inferred customer intent or dealer inventory.
Maturation/OOT and characteristic time	Top Anatomy; Temporal Grain respectively	Sibling research receipts only; no ruler or ownership absorption.

Owner map grounded in [I04, I16, I18, I19, I20, I21, I22, I23, I24, I25, I26, I27, I28, I29, I30]. Some implementation filenames are navigational anchors from the owner contracts, not independently rerun modules. The incumbent tech_confluence / Combo V1 semantics remain a controlled legacy comparison, not a replacement lifecycle engine or a new universal score. Its current full-module behavior must be checked before a benchmark is frozen; W0 does not authorize rebuilding it. [I04]

4.3 Identity namespace collision that must be resolved explicitly

There are two different “B1” labels in the estate: **Market OS B1** is the security-state product program, whereas **Prophet B1** is the canonical candidate-episode owner. Contracts and handoffs must always namespace them. Similarly, OpportunityCase is not security_state, and neither is a Radar event, an options campaign, a Plan, or a physical position. [I18, I19, I20, I21, I22]

A normalized OLI view therefore needs typed references, not one unqualified episode_id column. Its join must state the relation—observes, attaches-to, assesses, supersedes, re-arms, or belongs-to-plan—and the authority that established it. Cardinality must be explicit. One issuer can have several assessments; that does not authorize several duplicate physical positions or several simultaneous B1 episodes against the current one-active rule.

5. External research synthesis

5.1 What the market literature supports—and what it does not

Technical structure. Lo, Mamaysky and Wang found that systematically recognized technical patterns can alter conditional return distributions in their historical U.S. sample. That supports testing explicit, reproducible constructions rather than dismissing all technical geometry. It does not establish executable profit, causal early recognition, a particular breakout rule, or current Mastermind performance. [X01]

Momentum and proximity to highs. Jegadeesh and Titman's evidence concerns intermediate-horizon winner/loser continuation; George and Hwang show that proximity to the 52-week high contains information beyond familiar momentum measures. These results contradict a universal assumption that a stock near its high is automatically unattractive. They do not establish that every extended stock is a good entry, or that a monthly effect transfers to a five-minute signal. [X02, X03]

Volume and maturation. Lee and Swaminathan show that volume conditions momentum behavior and subsequent reversal patterns. The relevant research implication is interaction and lifecycle heterogeneity, not “high volume confirms everything.” Volume-derived transforms share information with the price move and must not be counted as independent witnesses. [X04]

Information diffusion and catalysts. Hong, Lim and Stein's findings are consistent with slower diffusion under lower analyst coverage and asymmetric treatment of bad news. The useful hypothesis is that catalyst age, qualified revisions and observed price incorporation may matter at different landmarks. A price move does not reveal investors' private information or prove a causal narrative. [X05]

Group effects. Moskowitz and Grinblatt find an important industry component in momentum. Zaremba and coauthors report breadth-related predictability across market and industry portfolios in many countries, with implementation-cost caveats. Both motivate strong group controls. Neither licenses today's theme memberships as historical facts or establishes a calibrated local-subtheme entry probability. [X06, X10]

Regime and tail risk. Daniel and Moskowitz document momentum crashes concentrated in particular market conditions. This supports conditional risk analysis, but their portfolio mechanism is not a universal sell rule for an individual long winner. A regime variable must improve the particular transition or holding decision after controls, not merely sound plausible during a drawdown. [X07]

Earnings and execution. Ng, Rusticus and Verdi connect post-earnings drift to transaction costs and show why apparent predictability need not equal realizable profit. Earnings continuation should be tested from a qualified, executable decision time, with expectation comparability, spread, delay, and the full original event population retained. [X08]

Options as a witness. Pan and Poteshman's predictive evidence uses a specialized measure of buyer-initiated opening option volume. Generic premium turnover lacks those semantics. The paper does not justify reading institutional intent from an unsigned feed, or treating estimated dealer positioning as observed inventory. [X09]

Compression, fakeouts and reclaim. This review did not establish a transferable, calibrated law for the commission's exact compression, failed-breakout or re-entry definitions. Those remain registered transition hypotheses. Pattern terminology, vendor chart annotations and selected examples are not substitutes for causal detection, full failure retention and out-of-sample comparison.

5.2 Why transitions and current landmarks are a defensible modeling problem

Landmark prediction estimates outcomes conditional on being eligible and observed at a specified decision time. It is a natural methodological precedent for asking what remains after development or deterioration without replacing the origin question. Landmarking 2.0 also illustrates the tradeoff between a relatively direct landmark approach and richer models of longitudinal covariates. These are statistical methods, not evidence of equity alpha. [X11, X29]

Competing-risk and multi-state methods distinguish mutually exclusive first events from subsequent transitions. For OLI, reaching invalidation first, producing confirmation first, and expiring are not interchangeable negatives. Cause-specific hazards and cumulative event probabilities answer different questions; a subdistribution model should not be relabeled an instantaneous cause probability. [X12, X13]

Hidden semi-Markov models explicitly represent duration, which could matter when an old armed setup differs from a newly armed one. Bayesian online change-point methods offer a causal run-length representation. Neither justifies hindsight-smoothed phase labels in a live decision. The first challenger should add elapsed duration to a transparent hazard model before adopting hidden states. [X21, X22]

5.3 Models, uncertainty and decisions

Gu, Kelly and Xiu provide finance-specific evidence that nonlinear interactions can improve risk-premium prediction in their setting. TFT provides a multi-horizon architecture separating static, historically observed, and genuinely known-future inputs. These support a later challenger ladder, not a presumption that sequence models beat a source-correct baseline for OLI. An earnings date known today may be a known-future input; future earnings content is not. [X23, X24]

Proper scoring rules reward honest predictive distributions. Censored-event scoring requires appropriate treatment of censoring; unresolved outcomes cannot simply be scored as failures or omitted. Adaptive conformal inference targets long-run coverage under distribution shift, while selective prediction studies the risk-coverage tradeoff. None of these automatically supplies conditional calibration for every stock, phase, regime or selected top-pick slice. [X14, X15, X27, X28]

POMDP theory is a useful conceptual language for decisions under partially observed state. Optimal stopping work shows how entry, exit, deadlines and costs can interact under specified stochastic models. Leung and Li's mean-reverting-spread assumptions are not a general law for equity leadership. OLI should first compare a small frozen set of admissible policies; an opaque reinforcement-learning trader is not the first deliverable. [X25, X26]

5.4 Search control is part of the hypothesis

White's Reality Check, Hansen's superior-predictive-ability test, and Sullivan, Timmermann and White's technical-rule study address the danger of selecting a favorable rule from many candidates. Harvey, Liu and Zhu emphasize the evidentiary consequences of extensive factor search. Backtest-overfitting diagnostics add useful selection-risk information, but do not replace chronological, point-in-time and prospective validation. OLI's trial budget must include label definitions, clocks, horizons, thresholds, policies, and feature families—not only final fitted models. [X16, X17, X18, X19, X20]

5.5 Research conclusion

Published evidence supports **conditional mechanisms worth testing**, not the full OLI hypothesis as an established fact. The scientifically strongest starting point is causal technical definitions plus price/relative-strength/volatility controls, coherent event targets, common-origin policy comparisons, and explicit incremental family tests. The strongest product claim available before those tests is improved interpretability, consistency and accountability—not improved investment returns.

6. Competitive product study

6.1 Scope and comparison law

Nine product families were reviewed through their current official documentation or developer-published descriptions: TradingView, MarketSurge, TrendSpider, Trade Ideas, Koyfin, TC2000, Bloomberg, LSEG Workspace/StarMine, and FactSet. This is a workflow comparison, not a purchased-product audit. **“Not established in the reviewed documentation” does not mean “the vendor cannot do it.”** No competitive accuracy or performance claim is independently validated here.

6.2 Workflow findings

Product	Documented workflow relevant to OLI	Implication for Mastermind
TradingView	Server-side strategy-alert copies; explicit differences between historical order-time display and real-time alerts; warnings about repainting	Reuse one accepted definition across replay and live delivery, but preserve the actual alert/model version and decision clock. [P01, P02, P03]
MarketSurge	Forming base-pattern recognition, Near Pivot/Breaking Out lists, RS/group context, chart buy/profit/loss zones and alerts	Discovery versus entry and phase-like chart guidance are not unique to Mastermind. The unmet claim must be calibrated remaining opportunity and accountable transitions, not colored labels. [P07]
TrendSpider	Multi-factor/multi-timeframe alerts; confirmation candles; shared visual scripts across scans/backtests/alerts	A reusable rule definition and explicit confirmation semantics are competitive baseline capabilities. [P04, P06, P19]
Trade Ideas Holly	Different risk modes with stops, targets, time exits, profit-save and risk-reduction rules; aggressive mode explicitly presentation-only	Separate entry from management and expose the governing policy. A more attractive displayed P&L under another policy is not proof that the original trade was good. [P08]
Koyfin	Watchlist/portfolio alerts and research context spanning news, filings, transcripts and financial data	Connect catalyst change to the user's attention set; do not make the user reconstruct the same company story across disconnected pages. [P09, P10]
TC2000	Shared conditions used in columns, filtering, alerts and grids; alert restart parameters	Alert restart is an operational mechanism, not scientifically accountable re-entry. Reuse expressions while preserving separate episode and prediction lineage. [P11, P12]
Bloomberg	Research management joins proprietary research and market context; PORT integrates positions, risk, performance and scenario analysis	Institutional workflow continuity is already a serious competitive standard. Mastermind should not claim uniqueness merely for joining research and portfolios. [P13, P14]
LSEG Workspace / StarMine	Signal Search, Watchlist Pulse and expectation analytics; SmartEstimates weight estimate recency and analyst accuracy	Catalyst and expectation state can be richer than calendar proximity. Licensed coverage, comparability and knowledge-time provenance remain prerequisites. [P15, P16]
FactSet	Research continuity, integration with watchlists/research tools and portfolio-aware internal research initiatives	Durable thesis history and research integration are competitive necessities, not a defensible moat by themselves. [P17, P18]

TrendSpider's documented Strategy Tester has important scope limits: a single selected timeframe, no repeated scale-in entries, and exits that close the whole position. Its multi-timeframe alert capabilities therefore cannot be assumed identical to every backtesting/position-management path. This is a useful warning for Mastermind: consistency must be proved across each actual consumer, not inferred from a shared product name. [P05]

6.3 The comparison questions answered

Phase or strength? MarketSurge provides phase-like pattern and zone guidance; TrendSpider and TC2000 support condition/state monitoring; Holly provides policy-specific trade management. Institutional suites connect research, signals and portfolio context. It would be inaccurate to characterize all competitors as static strength rankers. [P04, P07, P08, P11, P13, P14, P15, P16, P17, P18]

Discovery versus entry? Several products explicitly distinguish screening/watch conditions from entries or alerts. Mastermind should match this through its existing two-queue architecture, not claim the distinction as a new invention. [P01, P04, P06, P07]

Confirmation cost? Confirmation conditions are documented, especially by TrendSpider. The reviewed pages do not establish a common, calibrated economic cost-of-waiting analysis over original opportunities, including never-confirmers. That is a proposed research and product advantage, not a proven exclusive feature. [P19]

Remaining opportunity and management? Targets, stops, zones, estimates and risk modes provide relevant neighboring capabilities. The reviewed documentation does not establish the exact OLI bundle of origin-preserving current-landmark payoff distributions, remaining adverse-path risk, and distinct fresh-entry versus existing-position assessment. This bundle is the testable product thesis, not a claim that no institution has equivalent custom analytics. [P07, P08, P14, P16]

Reset and re-entry? Configurable alert restart is documented. That is not sufficient evidence of immutable failed-episode retention, independently accountable re-arm forecasts, or calibrated repeated-failure risk. Mastermind should make those laws observable in its product and evaluation record. [P12]

Probabilities or scores? The reviewed pages do not provide a uniform calibration protocol across products. A proprietary rating, target zone, simulated P&L or vendor-reported accuracy is not interchangeable with a horizon-specific calibrated event probability. Do not import vendor marketing numbers as Mastermind's benchmark.

6.4 Defensible advantage to pursue

The proposed advantage is **one consistent, source-qualified answer across discovery, entry, holding, reset and return visits, with immutable decision history and independently earned statistical claims**. Its ingredients exist in Mastermind, but the integrated capability is not yet proven. Competitive success should be measured through completed user tasks, correct transitions, source/degradation honesty, reduced reconstruction burden, and eventually prospective decision value—not a larger feature count.

7. Lifecycle ontology v1

7.1 Smallest useful representation

Proposal: retain each owner's native state and add a versioned, strategy-specific interpretation profile. Do not replace native enums with one global ordinal lifecycle. For the first technical profile, a small coarse stage set is sufficient: **FORMING, ARMED, TRIGGERED, CONFIRMED, TERMINAL**. A mapping is admitted only when the source can prove its predicates. Earnings and Cycle strategies may retain different native phases rather than being forced into a technical-breakout vocabulary.

This coarse profile is for interpretation and risk-set construction, not a second canonical state writer. A TOI native **EXTENDED** state remains visible as such; the profile must not erase it by displaying only **CONFIRMED**. Unsupported mappings are null with a reason. A state is economically useful only if it changes the forecast question, permitted action set, or next observation—not because it makes a historical chart look orderly.

Axis	Question	Proposed representation
Opportunity phase	What is developing under this species/strategy?	Native owner state + optional admitted coarse stage + transition receipt
Entry actionability	Can a fresh entry be considered under this exact strategy now?	B4/native verdict, reasons, quote basis, validity interval; no locally recomputed permission
Actual Plan/position	Has the user saved, entered, exited, or resolved anything?	Existing private Plan/position reference and native state
Management assessment	What needs attention for this existing plan?	Native policy assessment; observational flags and qualified forecasts, not an exit command
Thesis integrity	Which technical, catalyst, group or regime claim remains supported?	Typed component assessments, contradictions and unknowns; no blanket “price down = broken thesis”
Path and maturity	How far has price progressed, and what remains uncertain?	Measured path descriptors plus separately qualified forecast distributions
Time	Which observation, decision, prediction and expiry clocks apply?	Explicit clock and policy references, elapsed duration, remaining valid horizon
Evidence and data	How strong and current is each statement?	Native scientific status, source quality, missingness, calibration receipt and authority ceiling

The screen need not display eight equally prominent badges. Phase, current-entry status and Plan state belong in the first glance; the remaining axes explain them. This preserves the commission's four fundamental lifecycle distinctions while avoiding a Cartesian-product enum. [C00, I04, I20, I22, I30]

7.2 Classify the disputed words

FORMING is a species-qualified opportunity phase: necessary ingredients exist, but the lawful activation predicate is not satisfied. **ARMED/READY** means a known trigger and invalidation/expiry definition exist; it is not a buy instruction. **TRIGGERED** is both an observed event and the start of an activated technical phase. **CONFIRMED** means a registered additional evidence predicate was met; it must carry when that became known and what waiting cost was paid.

ENTER is a decision/action, not an asset state. **MANAGE/DELIVERING** describes a plan-specific task or assessment, not a new-entry recommendation. **EXTENDED/MATURE** should normally be path/maturity context; it can remain a native technical state where the owner defines it, but never universally means sell. **EXHAUSTED/DETERIORATING** requires a named observation or forecast target, not an impression from a large historical move.

RESET/PULLBACK is an observed development; **RE-ARM** is an owner-governed transition; **RE-ENTRY** is a new decision at an accountable landmark, sometimes on a new canonical episode. **INVALIDATED, RESOLVED, EXPIRED and RETRACTED** are terminal episode dispositions where supported by the native owner. **FAKEOUT/ FAILED** is a registered outcome or diagnostic: it need not be identical to economic-thesis invalidation.

DEAD MONEY is a horizon- and policy-specific time/opportunity-cost outcome, not a universal market phase. **INSUFFICIENT DATA/AMBIGUOUS** is an epistemic condition; **ABSTAIN** is the resulting policy choice when appropriate.

7.3 Allowed and forbidden transitions

From / development	Lawful treatment	Forbidden shortcut
Forming → Armed	Registered ingredients plus frozen trigger, invalidation and expiry become available	Armed because the model assigned a high uncalibrated score
Armed → Triggered	Native trigger event at its actual knowledge/executable clock	Backdating to the price extremum or unfinished higher-timeframe bar
Triggered → Confirmed	Named confirmation predicate with elapsed-time and price-cost receipt	Labeling only eventual winners confirmed and dropping failed starts
Any live phase → invalidated/expired	Owner transition with typed cause and time	Missing data or a red candle treated as economic invalidation
Active → reset observation	Append development; retain original origin and policy	Silently moving the original anchor or stop to rescue performance
Terminal → potential new opportunity	New admitted anchor/generation or other owner-approved economic episode; link prior history	Reopening a terminal episode by overwriting the failed outcome
Extended existing plan + unavailable new entry	Preserve both assessments	“Extended” automatically closes the position, or “Manage” opens another
Correction / retraction	Append lineage; retain as-run and corrected-as-known views	Repainting the originally published prediction as if corrected facts were then known

Direct forming-to-triggered observation can occur when a source first sees a gap through a threshold. It must be represented as first observed already triggered, with missing earlier observation history—not fictional evidence that the system previously armed the setup. Stage skipping is therefore permitted only as an honest observation history, not as backfilled monitoring success.

7.4 Transition explanation contract

Every displayed change must answer: **which owner changed which field, what newly admissible evidence caused it, what remained unchanged, what contradicted the change, and whether the implication is observational, predictive or actionable.** A text generator may summarize this structured diff. It may not invent a cause that the owner receipt does not support.

8. Episode identity and re-entry law

8.1 Current constraints are implementation law

Current Prophet B1 uses an owner-minted identity of the form

pe:<security_id>:<identity_epoch>:<anchor_token>:<generation>. The canonical anchor includes kind, time, price and basis; the data receipt provides provenance rather than a new identity on every refresh.

Ordinary source keys are immutable, unknown anchor kinds are suppressed, and a different anchor cannot replace an already active episode. The opening time cannot precede knowledge of the anchor. [I21, I22]

OLI must carry a **native-reference bundle**, not invent a global replacement episode key. Before B1 admission, a forming TOI occurrence or Radar nomination may still be shown under its native identity with

prophet_episode_ref = null and the exact absence reason. Absence of a B1 episode does not mean no market opportunity exists; it means that particular canonical strategy/population contract has not admitted it.

8.2 Same episode versus new episode

Same opportunity, new landmark: ordinary price development, a pullback still inside the registered thesis, additional confirmation, or an update that does not change the governing structural/economic anchor. Append an assessment; do not modify the original forecast.

New episode required: the prior owner episode has reached a terminal disposition and a materially new, admitted anchor or strategy-defined economic origin qualifies under the owner. Link rearm_of or an equivalent native relationship. Under current B1 code, merely resubmitting the same old anchor is not a new opening path. OLI must not evade this by perturbing a price or timestamp to create a different hash. [I22]

New thesis while an episode remains active: a catalyst or strategy change may justify an additional assessment, but current B1's one-active rule still applies. A requirement for genuinely simultaneous independent economic episodes needs an explicit B1 owner amendment and downstream migration study. OLI cannot grant that capability through a projection.

8.3 What persists and what can reset

Persist security/issuer lineage, original evidence and forecast, source-key custody, original entry/Plan policy, all failed starts, completed outcome, costs and prior failure count. A lawful new episode may establish a new anchor, trigger, invalidation, horizon, forecast and model version. It does not inherit the previous episode's favorable performance selectively or erase its unfavorable performance.

Re-arm qualification should specify geometry, duration and evidence rather than a universal cooling-off number. Examples for research include a completed pullback and reclaim of an owner-defined level, restoration of a predeclared risk/reward geometry, or a genuinely new source-qualified catalyst. These are candidate mechanisms, not admitted detectors. Repeated-failure count must be calculated from the then-known retained history, not only the successfully repaired episodes.

8.4 Hindsight-laundering attack tests

A valid implementation must reject a terminal-to-active overwrite, the same event with changed bytes, ticker/date surrogate IDs, an anchor known after the decision cut, and a failed episode later relabeled as a successful long-term thesis. It must distinguish an ordinary new observation from an explicit correction command.

Replaying a prefix must reproduce the originally published prefix even after later corrections arrive; a corrected historical view is a separately labeled query.

A user can deliberately open a new strategy plan after a loss. The UI must show that it is new, retain the earlier loss, and require an explicit new plan decision. No automatic plan transfer is implied.

9. Evidence-family matrix

9.1 Admission law

Every field needs subject identity, unit and basis, event/reference time, first-known/observed time, source revision, allowed use, missingness and dependency group. A provider's historical timestamp is not proof that Mastermind captured the observation then. For a decision cut C, all required knowledge clocks must be no later than C; a current-only source may support “known now” research without qualifying historical replay. Earnings D5 already embodies this distinction. [I16, I20, I29]

Family and owner	Mechanism / relevant transitions	Clock, dependency and missingness	Incremental experiment / ceiling
Technical primitives / TOI	Trend, compression, causal pivots, activation, reset, failure	Completed bars; source/session/adjustment basis. Most price transforms share a dependency group. History shortfall is unavailable, not bearish.	Add one registered geometry family to price/RS/volatility controls; descriptive until species evaluation earns more.
Relative strength / group owners	Leadership emergence, confirmation, continuation versus broad-market lift	Benchmark and peer membership known at cut; aligned sessions and currency basis. Do not count market RS, sector RS and momentum as three independent votes.	Price-only versus +market RS versus +group residual RS; same population and model budget.
Breadth / GMI	Participation versus concentration; leadership deterioration	Known membership denominator, eligible members and freshness. Current theme members cannot be backfilled. One leader driving an index is disclosed.	+breadth after own/group returns; compare raw and leave-subject-out breadth; test denominator stability.
Earnings / catalyst / Company Intelligence	New information, revisions, event continuation, invalidation	Fiscal period, units, basis, release and captured clocks, revision lineage. Unlicensed consensus remains absent.	Qualified facts/revisions beyond price and event-age controls; no automatic causal narrative or beat/miss inference.
Regime / rates / cross-asset owners	Conditional continuation, tail risk, time to realization	Only then-available readings; historical latest-revised series may fail as-observed use. Regime label must be causal.	Add predeclared interactions, not unrestricted state mining; preserve Phase 22's existing protocol.
Options / ThetaData / live flow	Incremental information, liquidity or event-risk witness	Trade and quote alignment, opening/closing availability, expiry, detector and campaign identity. Unsigned premium is not direction.	Matched time/coverage +price/catalyst controls, then measured quote-aware features; separate underlying versus option outcomes.

Family and owner	Mechanism / relevant transitions	Clock, dependency and missingness	Incremental experiment / ceiling
Stock Identity / native history	Instrument-specific behavior and source identity	Only prior admitted episodes; identity epoch and corporate actions explicit. Young/new names are a separate coverage class.	Hierarchical shrinkage versus no identity history; no memorized ticker lookup rescue.
Top Anatomy / analogs	Maturation and adverse-path descriptors	Frozen library tier and query-time eligibility; exclude future neighbors and protected OOT outcomes.	Equal-budget causal descriptors versus analog retrieval; retain null similarity/coverage; no new OOT ruler.
Liquidity / execution	Determines whether a favorable path is realizable	Quote age, spread, halts, session, participation and venue rights. L1 does not establish L2 queue position.	Cost and fill sensitivity on identical opportunities; executable permission only through the native owner.

Owner and data boundaries: [I04, I15, I16, I20, I23, I24, I27, I28, I29, I30]. Experiments in the last column are proposals, not accepted trial registrations.

9.2 Prevent false confluence

Attach a `dependence_group` to evidence at its native semantic family. Ten transformations of the same price history do not supply ten confirmations. Establish incremental value by removing a family, replacing it with a matched simpler proxy, and comparing the same model capacity and candidate population. A family can improve explanation or missingness awareness without improving forecasts; those are different acceptance claims.

Where optional evidence is absent, preserve the base eligible forecast only if its registered missingness policy permits that case. Do not fill absent options or catalysts with neutral zero. A model may use a predeclared availability indicator, but it must survive source-era and missingness-only controls; otherwise it may be learning the data pipeline rather than the market.

10. Transition-prediction design

10.1 Unit, risk set and executable time

Let i denote an owner-qualified opportunity/occurrence, s a registered strategy or species, l an accountable landmark, and F_l all admissible information at that cut. The predictive unit is (`native_ref`, `strategy/species version`, `landmark definition`, `decision cut`), not an unqualified ticker-day and not every refreshed UI row.

A forecast states its **risk set**. An armed-to-trigger question includes all armed cases, including those that invalidate or expire first. A triggered-to-confirmed question includes all lawful triggers, not just eventual confirmers. A holding forecast conditions on an actual or explicitly simulated eligible entry under a fixed policy. It cannot be compared directly with the original discovery cohort without accounting for that selection.

The first executable time is determined by source knowledge, completed-bar law, allowed session, processing latency and quote/fill availability. Event time and executable time are different fields. A late historical reconstruction cannot be labeled an early live prediction.

10.2 First-event targets

Transition question	Target and competing dispositions	Primary evaluation object
Forming → Armed/Trigger	Time to registered readiness or trigger; pre-trigger invalidation and expiry retained	Cumulative event incidence, time calibration, useful coverage and false preparation burden
Armed → Trigger	Lawful trigger before invalidation/expiry; gap-through and no-fill distinct	Activation distribution and executable activation coverage
Triggered → Confirmed	Confirmation before specified failure/invalidation or expiry	Confirmation incidence, delay, false-start rate and signed confirmation premium
Triggered/Confirmed → realized path	Net return/path distribution from actual executable landmark	Proper predictive score; paired policy value separately
Existing Plan → milestone/deterioration	Strategy-native realization, risk breach, thesis change and time stop	Remaining horizon risk, milestone timing, tail outcomes under fixed entry
Extended/reset → re-arm	Qualified reset and newly admissible geometry before terminal expiry	New accountable opportunity yield and repeated-failure risk
Any phase → invalidation	Technical, catalyst, group, regime causes, with unknown knowledge separate	Cause-specific event law and calibrated aggregate risk

“Fakeout” must be defined before labeling. It may be a diagnostic such as a return through the breakout boundary, while a deeper stop remains unhit. Such overlapping diagnostics cannot be added as if they were mutually exclusive first-event probabilities. When multiple causes coincide, the registered outcome ontology needs either an explicit joint category, a resolution rule supported by finer evidence, or an unresolved category. Arbitrary priority must not masquerade as observed chronology.

10.3 Transparent first model

A proposed initial model is a discrete-time competing-risk hazard with a no-event category. For mutually exclusive event types k and future interval u :

$$\begin{aligned}
 h_k(u \mid F_l) &= P(\text{first event is type } k \text{ in interval } u \mid \text{no prior event, } F_l) \\
 S(0) &= 1 \\
 S(u) &= S(u-1) \times [1 - \sum_k h_k(u \mid F_l)] \\
 F_k(H \mid F_l) &= \sum_{u=1..H} S(u-1) \times h_k(u \mid F_l)
 \end{aligned}$$

Use a multinomial parameterization or equivalent constrained construction so total first-event probability never exceeds one. The horizon forecast at landmark l cannot use realized future covariates. It uses current admissible covariates, registered duration effects, and only genuinely known-future calendar facts. New observations create a new forecast at a later landmark; they do not retroactively improve the old prediction. Competing-risk literature supplies the methodological basis, while these targets and interfaces are OLI design proposals. [X12, X13]

10.4 Model ladder and stop rules

- 1. Deterministic owner states and base rates.** Establish causal correctness, label accounting and useful display before fitting.

2. **Regularized logistic/discrete-hazard/additive models.** Include elapsed duration, simple price/RS/volatility, and a small number of registered interactions.
3. **Hierarchical partial pooling and calibrated boosting.** Admit only if sparse groups or nonlinear interactions provide measured incremental benefit with the same information and search budget.
4. **Multi-state, latent-duration, change-point, sequence or graph challengers.** Require a specific residual failure of simpler models, sufficient independent evidence, faithful explanation tests, and bounded compute/operational cost.

The ladder is a proposed selection procedure, not a claim that every rung is necessary. Hidden-state models must use filtered probabilities at decision time; retrospective smoothing is a diagnostic view only. An attention plot is not proof of causal importance. A complex model that adds no economically material, robust benefit is rejected even if its aggregate fit looks marginally better.

11. Remaining-opportunity framework

11.1 The object is a conditional distribution, not a score

Define remaining opportunity as **the distribution of feasible future reward, adverse path, invalidation and time from a specified current executable landmark, under a specified strategy and information set.** Keep the asset-path forecast separate from any policy-dependent payoff distribution. Neither is “distance to analyst target,” inverse RSI, or the percentage of a future top that has not yet happened.

The proposed output vector has six parts: future net-return quantiles; adverse-excursion quantiles; favorable-excursion quantiles; target/invalidation/expiry incidence; time-to-realization distribution; and uncertainty/coverage. An unqualified head is null, not a model-looking number.

11.2 Descriptors versus forecasts

Measured as of the landmark	Estimated only after qualification
Return since immutable origin	Distribution of future return from the current executable quote
Distance from original trigger and current invalidation	Probability of invalidation within the declared remaining horizon
Extension normalized by an explicitly frozen or current volatility scale	Remaining favorable/adverse excursion distribution
Giveback from the peak observed so far	Probability of continued leadership versus deterioration
Signed price premium paid for confirmation	Expected economic value of waiting versus entering now
Elapsed sessions and remaining registered expiry	Time to realization, time in dead money and uncertainty

Call the first group **progress since origin / extension**, not “percent of total opportunity consumed,” unless the denominator is an explicit frozen plan objective and labeled as such. The eventual total move is unknown at the landmark. Recomputing a historical origin-volatility scale with later data is also forbidden.

11.3 Price and payoff basis

For a long-only executable forecast, an illustrative accounting convention is:

```
net_return(l, t) = (exit_bid(t) + eligible_distributions(l,t) - entry_ask(l))
                  / entry_ask(l) - non_spread_costs(l,t)
```

This is an accounting specification, not a fitted model. The price/adjustment and distribution convention must avoid double-counting dividends or corporate actions. A midpoint-plus-spread model is an alternative, not an extra spread charge on top of bid/ask accounting. Borrow, financing, short-sale constraints and option Greeks/expiry require their own qualified contracts before extending the first long-only equity slice.

MFE is an upper-path excursion diagnostic; a strategy does not capture the maximum merely because it was observed later. MAE from entry is not running-peak drawdown. Report each with its own denominator and sampling resolution. A future high/low inside a coarse bar may establish an excursion bound without establishing an executable sequence or fill.

11.4 Confirmation and waiting

A signed confirmation premium compares the first lawful executable price with the later confirmation price for the same opportunity. It can be negative: waiting can improve price or avoid a failed entry. The actual value-of-waiting experiment must retain cases that never confirm and cash while waiting, rather than compare successful early trades with successful late trades from different populations.

Use two separate rulers. The **forecast ruler** asks about a stated horizon from each landmark. The **policy ruler** compares complete policies over a common origin-to-expiry accounting window. Giving every late entry a fresh full holding window can change the economic question and should be a separate registered study, not an unnoticed advantage.

11.5 Four different judgments

A strong company can have a weak current entry. An excellent original call can have little remaining opportunity. An unattractive fresh entry can coexist with an appropriate existing hold because position constraints, costs and strategy differ. However, the underlying future price path from today is not improved by having bought earlier. Original P&L is historical attribution, not a reason to assign a more favorable future asset forecast to an existing holder. Preserve that distinction to avoid encoding sunk-cost reasoning into “management intelligence.”

12. Management-intelligence framework

12.1 Freeze the governing contract at entry

Freeze the actual decision/fill reference, strategy/version, entry source basis, original thesis, invalidation law, expiry, sizing/risk owner reference, initial forecast and applicable management policy. Subsequent evidence may update forecasts and attention flags, but it cannot rewrite what the original trade promised. A policy amendment is an explicit new decision with lineage, not a silently changed stop or horizon. This extends the strategy-native hold law already present in R6. [120]

12.2 What management should observe

Track technical invalidation separately from ordinary volatility; catalyst facts separately from price interpretation; peer/breadth deterioration separately from the subject's return; and macro context separately from the plan's authorized response. Measure progress toward the registered thesis, time remaining, current adverse-path risk, liquidity changes, and source quality. A stale source means the assessment may be unavailable; it does not certify either safety or failure.

Hidden fragility should be a registered conditional-risk hypothesis, not an evocative label attached to every losing chart. Phase 22's rising-real-yield and weak equal-weight breadth interaction remains that existing owner's prospective question. OLI must neither retune it after viewing the earlier forensic set nor make it a live veto without promotion. [I23]

12.3 Entry and management must be experimentally separated

Use a staged factorial design. First vary entry policy while holding management fixed. Next vary management while holding the entry cohort and execution rule fixed. Only then test a separately registered entry-by-management interaction. This can distinguish a late-entry problem from a deterioration-during-hold problem. An apparent "better management" system that quietly selects easier entries has not answered the question.

The first management product should support **observe, review, and explain**. Suggested review conditions can include a source-qualified thesis change or approaching expiry. They must not automatically trim, sell, add, move stops or re-enter. Any future action policy requires its own feasible portfolio/risk evaluation, consumer-specific authority and rollback.

12.4 Time stops and opportunity cost

A time stop is strategy-native: failure to achieve a registered milestone by a defined deadline, not an ex-post observation that another stock did better. Evaluate it against the same opportunity set with cash and feasible alternative-use assumptions clearly separated. Equal-opportunity simulations are not investable portfolio performance; overlapping positions and capital constraints require the existing portfolio owner.

13. Characteristic horizon and regime integration

13.1 Six clocks, not one timeframe dropdown

Clock	Role	Admission requirement
Observation grain	Raw bar/event sampling	Exact vendor, session, exchange close and adjustment semantics
Anchor/session grammar	Where windows start and end	Named calendar/version; early closes and cross-market sessions explicit
Filter memory	How much past information affects a feature	Fixed causal recipe; no hidden dependence on future bars
Prediction horizon	Future quantity being estimated	Registered endpoint and clock; no off-horizon verdict
Episode/plan expiry	When the original opportunity or mandate ends	Native strategy law, preserved across later forecasts

Clock	Role	Admission requirement
Management/review cadence	When a new assessment is appended	Predeclared scheduled or event-driven landmarks, deduplicated and accountable

TOI owns broad data/clock admission, and Temporal Grain retains its characteristic-time study. A smoother chart or attractive motivating example does not establish a better decision clock. Derman's intrinsic-time formulation is a conceptual precedent, not validation of a stock-specific best timeframe. [I07, I25, I26, X30]

13.2 Proposed adaptive-horizon experiment

Start with the incumbent fixed-clock policy. Add a causal duration/volatility descriptor without changing the horizon. Only then test a small predeclared clock-selection rule derived on permitted development data and frozen before validation. Compare it against fixed-clock and equal-compute alternatives, preserving the same origin population and information availability. Instrument-disjoint and regime-separated tests should follow the Temporal Grain owner's accepted sequence.

Potential clocks include completed sessions, elapsed trading minutes, catalyst-relative time and a source-qualified volatility/activity clock. A forecast horizon, lifecycle expiry and management cadence may differ. Faster monitoring does not justify a shorter original thesis or more independent observations.

13.3 Cross-market boundary

The first empirical vertical should remain within an actually qualified market/sleeve/session. U.S. RTH evidence must not silently authorize China, Hong Kong or Canada. Each market needs its own exchange calendar, sessions, corporate-action basis, instrument and currency identity, liquidity/price-limit rules where relevant, rights, and historical eligibility. The architecture can share interfaces while its admitted population remains narrow. A unified product is not a universal scientific permission.

14. Research and statistical protocol

14.1 Three distinct acceptance claims

Semantic/engineering value: can the system reproduce source truth, preserve identity and explain changes consistently? **Predictive value:** do the new heads forecast the registered outcomes better than strong same-information baselines? **Decision value:** does a feasible policy using them improve a declared economic objective after costs, waiting, failures and coverage? A pass in one category does not imply a pass in the others.

14.2 Population and denominator law

Freeze the population rule before outcomes: market, security types, historical eligibility, source era, session, species, anchor admissibility, and observation-start rule. Retain every nominated/forming/armed/triggered opportunity and its disposition, including suppressed, passed-over, unpriced, stale, never-triggered, never-confirmed, expired and failed cases. Do not backfill current members into a historical universe.

TOI W0 starts its forward evidence episode at trigger. Therefore **pre-trigger research has a real denominator requirement:** determine which existing nomination, near-miss, suppression or evaluation owner can retain the first qualifying forming observation. Extend that owner only where its current contract lacks the necessary reference and timestamps. Do not fabricate B1 episodes for unanchored candidates or introduce a separate OLI event store. [I04, I22]

Report the accounting identity at each step: observed inputs = mapped + suppressed; mapped = qualified + unqualified under named reasons; target-eligible = resolved + administratively unfinished + unresolved/invalid measurement dispositions. These are separate denominators, not filters quietly applied to the final win rate.

14.3 Registered landmarks

Choose a bounded landmark schedule before outcomes, such as first forming observation, first lawful trigger, registered confirmation, fixed strategy review intervals, and independently defined material changes. Deduplicate identical assessment keys. Event-driven landmarks must have a causal trigger independent of the future outcome. Store the original and every subsequent prediction through the existing prediction/outcome owner.

A price path that generated 100 five-minute refreshes remains one dependent opportunity history, not 100 independent successes. Explain whether estimates are episode-weighted, date-weighted, landmark-weighted or user-decision-weighted. The default policy comparison should prevent long-lived episodes from winning simply by contributing more rows.

14.4 Chronological development and validation

The proposed development design has an outer chronological evaluation split and inner chronological training/tuning/calibration partitions. Split whole decision dates, then enforce episode and label-information intervals. Purge training examples whose outcome interval overlaps a validation/test information boundary. Express embargo in the owner's trading-session clock, not blindly in row counts. Training labels must have matured before fitting.

Feature scaling, imputation, universe selection, interactions, threshold tuning, model selection and calibrator fitting all occur inside the appropriate training partition. Freeze the final candidate and calibrator before the untouched test. After the historical test is used, it becomes development evidence for a later iteration; the next confirmation needs a new unexposed period or prospective epoch.

Do not invent fixed historical start/end dates before the data owner establishes usable source eras. The registration must fill exact dates, horizons and admissible source-generation ranges before a target outcome read. A default 70/15/15 row split is not an acceptable substitute.

14.5 Baselines and ablations

The mandatory controls are the exact current static Buy/Wait/Extended implementation, TOI's two queues, existing Prophet admission/entry policy, simple price/RS/volatility models, same-information transparent logistic/hazard models, deterministic rule policies, and a useful abstain/cash reference. Freeze versions and actual input populations; do not weaken the incumbent retrospectively.

Separate **richer information** from **richer modeling** using a crossed comparison: baseline information with simple and complex models, then added family information with both. Run leave-family-out, matched-proxy, source-era, missingness-only, and current-only versus PIT-admitted sensitivity checks. A model cannot claim incremental catalyst value merely because its catalyst cohort has better data or larger, more liquid firms.

14.6 Endpoints and economically meaningful gates

For transition forecasts, predeclare a proper event/distribution loss plus calibration and coverage diagnostics. For an entry policy, the proposed primary is a **date-weighted paired full-opportunity net outcome difference**, measured over the common origin-to-expiry window with declared cash while waiting. Supporting views include episode-weighted and fill-conditional outcomes, severe-loss incidence, retained large-winner contribution, MAE/MFE, turnover, time to payoff, and cost sensitivity. They are not interchangeable primary endpoints. [120]

A formal promotion decision must precommit: economic minimum improvement δ , harmful-endpoint non-inferiority margin m , minimum useful coverage c , acceptable uncertainty, exact multiplicity family, and the formal look schedule. Derive these from the consumer's loss budget and permitted pilot variance/power analysis, not from protected outcomes. This report does not invent universal numerical thresholds and falsely label the study registered. Filling these numbers, with owner approval before outcomes, is an explicit wave acceptance deliverable.

Existing study gates remain binding. Phase 22's months/episodes/effective-name floors and Top Anatomy's maturity and one-look law cannot be replaced with a convenient OLI sample threshold. A large row count cannot compensate for insufficient independent dates, names or adverse events. [I23, I24]

14.7 Dependence and uncertainty

Report rows, native occurrences, B1 episodes, issuer counts, distinct decision dates/months, event counts, and concentration-based effective-N separately. Use a prespecified date-block or other owner-approved dependent-data resampling scheme that preserves entire episode/landmark histories and shared market shocks. Assess issuer concentration and overlapping holding intervals explicitly. The exact block length is selected on permitted development evidence or fixed by the incumbent ruler, never optimized to make a protected result significant.

Sparse regime/cause cells should use shrinkage or remain unestimated. A narrow confidence interval from incorrectly independent repeated rows is a defect, not high certainty. Informative missingness, trading halts, delisting and source failures require separate dispositions and sensitivity analyses, not a casual independent-censoring assumption.

14.8 Censoring, ambiguity and execution

Administrative horizon immaturity can be right-censored under a declared event-model convention. A delisting is an economically meaningful event with a recovery/return treatment, not automatically harmless censoring. A data gap is measurement uncertainty, not a zero return. A halt can prevent execution even when a model forecasts a favorable eventual path. If a target and invalidation occur in one coarse OHLC bar, chronology is unresolved unless qualified finer data resolves it; an adverse-first convention is a sensitivity bound, not an observed fact.

Replay must use the appropriate quote side or an explicit spread/slippage model, session permissions, latency, limit-order fill rules and capacity assumptions. No fill is a policy outcome. Unknown prices are not cash. Current L1 quotes do not prove historical queue execution or L2 capacity. Preserve the exact adjustment convention across features, targets and execution accounting. [I07, I20]

14.9 Calibration and changing risk

Report calibration at declared horizons and on meaningful selected slices, not only pooled across all rows. Include event-probability reliability, proper scores, interval coverage and width, and risk-coverage behavior where abstention is used. Confidence in a source, probability of an outcome and model uncertainty are separate quantities. A score becomes a probability only with a defined outcome, population, horizon, model and calibration receipt.

Use proper scoring and censoring-aware methodology as applicable; do not claim an average coverage guarantee provides conditional calibration in a thin regime or among today's top picks. Monitor source drift, missingness, covariate drift and realized forecast performance separately. Retraining is an owner-governed release, not automatic production learning from recent P&L. [X14, X15, X27, X28]

14.10 Multiple testing and prospective read law

Register the entire construction budget: state definitions, anchor/reset rules, clocks, horizons, features, interactions, models, calibrators, thresholds and management policies. Identify confirmatory versus exploratory families. Use the existing owner's multiplicity procedure; for a new small confirmatory family, a prespecified familywise-error method is preferable to retroactively choosing the most permissive correction. Search-aware comparisons such as Reality Check/SPA can supplement the analysis when their assumptions and candidate universe match the question. They do not erase undeclared exploration. [X16, X17, X18, X19, X20]

Prospective accrual starts only after a durable epoch receipt, source/transport admission, exact model/config hash, TrialLedger registration and one real end-to-end event prove the path. Counts and missingness may be inspected according to the ruler; protected outcomes may not be peeked at early. A failed or inconclusive result is retained. No horizon, threshold, subgroup or model "rescue" may be added to the same confirmatory epoch after seeing the answer.

15. Data capability matrix

15.1 Qualification is claim-specific

"Data exists" is not one capability. A dataset can support current display yet fail as-observed historical research; a regular-session test can pass while early-close semantics fail; measured options trades can exist while their durable outcome publication is not accepted. Each consumer must request an explicit use class: current observation, retrospective description, PIT replay, prospective prediction, or policy/execution evaluation.

Data capability	Current disposition from this census	Safe use / next qualification
Broad Weekly/Daily/4H technical substrate	PARTIAL / NOT ADMITTED under #7094	No broad technical outcome study. Prove exchange-close-aware aggregation, same-basis family and historical eligibility. [I07, I08]
Regular-session 4H parity sample	Reported 15/15 pass	Bounded sample evidence only; no extrapolation to all sessions/names. [I07]
Early-close 4H semantics	Reported 5/5 fail	Actual exchange close must determine bar endpoints and partial final bar handling. [I07]
Provider-hourly fallback where 5m absent	Insufficient for exact 09:30 parity	Label unsupported, or qualify a lawful source family; do not relabel 10:00 rows as 09:30-origin bars. [I07]
Historical universe, delisted/new names and eligibility	Partial / not fully certified	Freeze PIT inclusion, exclusion reasons, corporate actions and coverage by era. [I07, I20]
Radar forward event path	Existing owner, acceptance/transport gates	Consume only accepted epochs; fix R2/local intake under the incumbent before interpreting empty intake. [I12, I15]
Stock Identity behavioral history	Owner/identity boundary known; current behavior coverage not established here	Do not infer a complete historical behavioral library from B1 identity support. Qualify the then-known native history before adding behavioral features. [I20, I21, I22]

Data capability	Current disposition from this census	Safe use / next qualification
Canonical Prophet episodes	Core implemented; narrow anchor vocabulary	Exact B1 read/generation with explicit null for non-admitted anchor populations. Current live population coverage not recertified. [I21, I22]
Current B4 entry availability	Existing strategy-native contract/implementation referenced by R6	Read exact owner verdict and evidence; consumer-specific served acceptance still required. Do not infer it from candidate rank. [I20]
Catalyst/earnings expectations	Narrow admitted facts; broader repair/coverage partial	Revision-aware allowlisted evidence and typed absence; no unlicensed consensus or stale-as-current substitution. [I15, I29]
Historical real-yield first-known	Not reconstructable from latest-revised rows alone	Use qualified forward known-at/vintage receipts; preserve Phase 22 start law. [I23]
GMI membership and ThemeState	Substrate exists; broader admission staged	Retain membership vintage, rights and eligible denominator; no all-history/current-membership shortcut. [I27]
Options measured microstructure	Real evidence exists; publication/runtime acceptance separate	Descriptive witness with exact trade/quote clocks. Do not redo the accepted natural evidence hunt. [I28]
Options historical associations	Research package complete, not promoted	Retrospective research context only; no directional confidence or options-profitability inference. [I28]
Top Anatomy OOT	Naturally accruing, no new verdict	Preserve frozen maturity/one-look protocol; historical descriptors retain original tier. [I24]
Temporal adaptive clocks	Specification merged; usefulness not established here	Consume only future admitted sibling findings; fixed qualified clocks first. [I25, I26]
Whole-universe cross-surface live OLI	Not established / not built as an integrated contract	Prove one bounded subject/strategy through real owners, transport, auth and renderer before expansion. [I16, I17, I18, I19, I20]

15.2 Evidence not obtained in this run

This report did not inspect every private raw dataset, rerun W1/W2 tests, verify all current hosted checks, prove the latest production B1/B3/B4 population, resolve every outstanding GMI/catalyst carrier, or execute authenticated Terminal/Macro browser flows. The complete current D0_NOT_REBUILD.md and all primitive/registry modules were not exhaustively re-audited; binding no-duplicate and killed-construction constraints were recovered through the accepted owner documents and carriers. The next execution gate must read the exact relevant current kill entries and affected modules before implementation or trial registration.

A bounded host-side read attempt was blocked; it was not retried through a workaround. Native source and organizational reads provided the census evidence listed above. These limits are material: they prevent a blanket claim that all data or runtime paths are qualified, but do not prevent an evidence-grounded ownership decision and a precise recovery plan.

16. System architecture and derived contract

16.1 Cross-owner flow

```

Data / identity / sessions / source revisions
|
+--> TOI native primitives, occurrences and technical facts
+--> Radar native tactical events -----+
+--> Earnings / themes / options / regime facts      |
                                                    v
Prophet B1 canonical episodes <--- qualified native attachments only
|
|               (not ticker joins or new aliases)
+--> B3 candidate phase/maturity projection
+--> B4 strategy-specific present entry availability

All admitted owner facts + exact knowledge cuts + owner receipts
|
v
Existing Alpha K3-E Opportunity Evidence Vector / governed OLI profile
|
+--> Existing prediction owner: immutable origin + appended landmarks
|
|       -> TrialLedger / QLedger / Evaluation OS / Species adjudication
|
v
Existing Market OS security_state / opportunity-context projection
|
+--> Prophet Candidates and security detail
+--> Terminal chart / Tactical / Radar consumers
+--> Existing private Plan / Portfolio context join
+--> Existing alert and change-delivery owners

```

This is the recommended composition, not an installed new pipeline. Forecast storage and outcome grading use existing owners; the OEV join remains a derived view. K5 OpportunityCase remains subject to its own upstream dependencies. A prose synthesis cannot become a hidden numeric ranking input. [\[I16, I17, I18, I19, I20, I22, I30\]](#)

16.2 Proposed field profile—not a registered schema

The following is a **logical extension proposal for the existing OEV/security-state seam**. It is not authorization to introduce a new durable object or silently widen vector.v1.

```

profile_version           # governed existing-owner extension
subject_ref               # native security/issuer/listing identity
native_opportunity_refs[]  # schema + owner + ID + generation
prophet_episode_ref?      # only an owner-issued B1 identity
strategy_ref / species_ref # separate when not the same concept
plan_context_ref?         # private, not in the public shared payload

origin_ref / first_prediction_ref? # never replaced by a later assessment
landmark_ref / landmark_definition
native_event_times / knowledge_cuts # per source, not one synthetic clock
source_generation_refs / correction_refs
computed_at               # assembly time, not market-data freshness

phase.native / phase.profile_mapping?
entry_availability.owner_ref / verdict / reasons
management_assessment_ref? # not an order or position mutation
thesis_component_refs[]    # support / contradiction / unknown
horizon_ref / expiry_ref / cadence_ref

```

```

trigger_ref / invalidation_refs[] / rearm_condition_ref?
path_descriptors[]           # units, price basis and origin scale
forecast_refs[]              # target, population, horizon, model,
                             # calibration, intervals, evidence class
supporting_evidence[] / contradicting_evidence[]
missing_evidence[] / required_slot_status

source_freshness_by_slot / dominant_degradation
scientific_status_refs / authority_receipts
oli_authority_flags          # all false at birth

```

Do not insert LLM-generated confidence, conviction, universal scores, or fabricated neutral placeholders. Some incumbents have closed vocabularies and forbid particular keys; compatibility must be demonstrated by an explicit schema/version amendment and existing validation tests before adoption. [I16, I29]

16.3 Durable history: minimum necessary delta

Existing owners already hold native episode events, source revisions, Plan decisions and prediction/grade records. The likely missing extension is a **qualified relationship between a registered landmark prediction and its native opportunity, strategy, evidence snapshot and policy**. First census the current QLedger/arena/local-ledger interfaces and extend the responsible owner only where the relation is absent. No new prediction ledger is justified by this report.

A derived projection can have a content digest or cache key composed from profile version and owner generation references. That is not a replacement episode identity or a new system of record. A projection cache must be evictable and reproducible. Durable scientific truth remains the immutable owner record and evidence snapshot, not whichever card was last rendered.

16.4 Recompute and freshness

Recompute the display when an admitted owner generation changes. Append a new forecast only at a registered landmark, not on every UI refresh or quote tick. A forecast tied to an older executable quote must disclose that basis; displaying a fresh quote beside it does not silently reprice or recalibrate the prediction.

Each source has its own freshness and knowledge clock. A slow-moving earnings fact may remain valid longer than an intraday quote. The composer must not use the minimum or maximum timestamp as a false universal “live” badge. Required-slot failure blocks the dependent claim; optional absence remains visible and does not automatically suppress every unrelated fact.

Last-good fallback must preserve a coherent owner generation and display its age. It must not combine old strategy permission with a new price as though the original permission still applies. Inconsistency, unknown identity, unknown schema, future knowledge and invalid correction lineage fail closed for the affected decision claim.

16.5 Authority, privacy and delivery

All OLI-originated authority flags begin false. A separately accepted B4 entry verdict may be referenced and rendered as **B4's verdict**, not reproduced as new OLI permission. An OLI forecast cannot turn an owner veto into a pass. Statistical acceptance is not automatic ranking, sizing, alert, order, or exit authority; each consumer needs a distinct promotion receipt.

The public/shared projection must not carry private Plan holdings, quantities or user-specific risk assumptions. Join that context through the existing authenticated private owner. Existing cache keys and transport paths must preserve subject identity, account boundaries and generation consistency. Terminal and Macro should render the same canonical semantics, not independently infer state from price arrays.

17. Product architecture

17.1 The primary journey

A user opens the desk and sees **what changed**, then a forming or triggered opportunity, then a concise security view explaining what can be considered now. Saving goes through the existing canonical watchlist/plan path. An actual position appears only after the native position owner establishes it. Alerts bring the user back when a named condition changes. The return visit shows the difference from the prior accountable landmark, not a rewritten story.

The first glance should answer: phase; strategy and horizon; present-entry status; actual Plan/position status; one meaningful support; one meaningful contradiction; and next observable condition. Trigger, invalidation, expiry and source quality should be available without opening a research essay. Probabilistic path bands appear only when that specific head is qualified.

17.2 Surface responsibilities

Surface	Primary job	Must not do
Prophet Candidates	Comprehensive population plus selective strategy views; separate forming/armed from triggered opportunities	Hide failed/non-admitted cases from research accounting or collapse all sleeves into a shortlist
Prophet Plans	Monitor a saved strategy and actual user decisions	Infer a fill from a save, transfer a failed plan into a new thesis silently, or treat "Manage" as "Buy"
Technical Opportunity view	Explain native geometry, activation, confirmation cost and reset conditions	Mint alternative primitives, species or B1 episodes
Terminal security/chart	Show causal annotations and current context beside price	Backpaint known-later pivots as earlier calls or recompute a competing lifecycle from chart pixels
Live Entry Radar / Tactical	Surface existing native tactical events and their qualified context	Build another scanner, spool, event ID or evaluator
Theme/subtheme view	Explain member participation, concentration and opportunity mix	Treat missing members as weak breadth or add another theme-state authority
Alerts / return visit	Deliver a named, source-backed change at the right landmark	Promote descriptive change alerts into trade signals or equate alert restart with scientific re-entry
Evidence detail	Reveal source, clocks, uncertainty, sample size, contradictions and old forecasts	Conceal partial coverage behind a confident summary

These are proposed responsibilities composed over the existing owners, not a new navigation rewrite. Preserve the current Prophet dense screener and selective top-picks views; both consume the same admitted context rather than creating separate truth.

17.3 A concrete card hierarchy

Header: security identity, selected strategy, independent holding horizon, and source state. **Decision row:** native phase; native fresh-entry verdict; actual Plan status. **Geometry row:** trigger, invalidation, expiry and quote basis. **Evidence row:** strongest support and strongest unresolved contradiction. **Change row:** previous landmark, what changed, and next condition. **Research drawer:** origin/current forecasts, uncertainty, effective sample and prospective evidence class.

Use progressive disclosure. Do not place every source timestamp, method name and research disclaimer in the first card. Conversely, do not remove useful functionality to make the design sparse: detailed tables, chart context and evidence remain one level away. Colors should supplement text and icons rather than being the only means of distinguishing buy/hold/invalidated states.

17.4 Contradiction examples

A hypothetical security can display **confirmed leader / fresh entry unavailable / existing plan under review** without contradiction in the ontology. Another can display **attractive technical reset / catalyst deterioration**. A third can show **breakout confirmed / peer participation weak**. These are synthetic examples of the proposed interface, not live security recommendations or validated signals.

The explanation must state which axis differs and why. It should never “resolve” a real contradiction by averaging all evidence into one confidence score. A long-term thesis can remain intact while a short-term entry is unattractive; the displayed strategy and horizon make that distinction legible.

17.5 Required degraded and correction states

Distinguish: no opportunity found with healthy coverage; source unavailable; transport unavailable; event exists but consumer not accepted; episode anchor not admitted; stale last-good evidence; optional family not covered; model unvalidated; forecast unavailable at this landmark; rights-blocked source; expired plan; and corrected historical evidence. The Radar intake defect makes the first three distinctions especially important. [I15]

Show old and new prediction cuts together when useful. A correction should explain what the system knew then and what is known now. A failed first episode followed by a successful re-entry remains two accountable decisions.

17.6 Browser and API acceptance

Prove one real eligible subject/strategy through the existing Macro/Terminal gateway and actual access-control path. Test source freshness, schema/version equality, null handling, no horizontal overflow at desktop/tablet/mobile widths, keyboard navigation, focus recovery, light/dark readability, and bilingual copy where the existing surface supports it. Screenshot fixtures prove rendering, not the live transport path; both receipts are needed for production acceptance.

API and UI consumers must receive the same phase, reasons, timestamps and authority. A snapshot test alone is insufficient: include state-change, expired, missing, correction and replay cases. Preserve existing accepted UI work and targeted incumbents; OLI is not permission for another global dashboard rewrite.

18. Build dependency graph and waves

18.1 Two tracks with one evidence boundary

```

R0 Recovery / ownership reconciliation
|
+--> C0 Contract, identity and denominator design
|
|      |
|      +--> P0 Honest observational product on already-admitted facts
|      |
|      +--> S0 Registered lifecycle targets and first tournament
|
+--> D0 Existing W1 / W2-0 closure and bounded W2 repair
|
|      +---[with S0]--> H0 Historical/PIT evaluation
|
Existing Radar/private transport + accepted epoch plumbing -----+
                                                                    v
                                                                    F0 Forward shadow
                                                                    |
M0 Management and current landmarks <-----+
E0 Reset / re-entry <-----+
X0 Incremental cross-domain evidence <-----+
|
P1 Unified qualified Prophet / Terminal experience
|
A0 Consumer-specific authority adjudication

```

The graph expresses evidence dependencies, not automatic dispatch. C0/P0 can make useful progress while data/science remain held, provided they use admitted facts and do not open prohibited outcomes. M0/E0/X0 definitions and fixtures can be prepared early, but their empirical claims require the relevant qualified data, registration and evaluation path. Nothing in this graph absorbs the live #784 execution.

18.2 R0 – Recovery and ownership acceptance

Accountable owner: current semantic/TOI principal, coordinating incumbent owners; no new assignment is made here. **Inputs:** this census, current protected procedure, live carrier heads, owner decisions and actual active-session receipts. **Outputs:** accepted ownership decision; exact owner/carrier/status map; linked existing PRs in Agent OS/Linear; explicit gaps and next actions. **Dependency:** none beyond current-source and custody gates.

Gates/tests: distinguish source, runtime, product and statistical status; verify that #7107/#7094 are not duplicated, #784 remains active, Top's wait survives, and repaired B1 allowlist is not recommissioned. **Product:** internal continuity only. **Authority:** records only. **Rollback:** revert only the bounded records amendment, never accepted implementation or incumbent work. Reconciliation writes that overlap #7094's existing Agent OS change belong with that carrier or an explicitly coordinated owner, not a parallel conflicting patch.

18.3 D0 – Incumbent evidence and data closure

Owners: W1 and W2 incumbents, reviewed by their existing accepting authority. **Inputs:** #7107 exact head plus protected dirty amendments; #7094 exact head, parity receipts, session and basis owners. **Outputs:** accepted W1 evidence catalog and conservative counts; accepted W2-0 archaeology record; separately qualified bounded W2 source family and coverage manifest. **Research gate:** no market-outcome research before the particular source family is admitted.

Tests: generated catalog-ID validator; intended duplicate-membership guard; numeric bandwidth binding; frozen/latest Donchian distinction; count reconciliation; actual early-close endpoints; exact regular/early session parity; raw/adjusted/corporate-action consistency; historical inclusion and absence denominators; current-base integration and hosted exact-head checks. **Product:** data-status improvements, not new forecasts. **Authority:** no blanket signal promotion. **Rollback:** withhold the rejected family/version and preserve last admitted source; never repair acceptance by weakening tests or restamping historical knowledge.

18.4 C0 – Identity, ontology and derived contract

Owners: Alpha K3 semantics with TOI, Prophet B1/B3/B4, Market OS and Evaluation sign-off. **Inputs:** current native schemas and allowed extension points. **Outputs:** versioned mapping profile, native-reference relation law, pre-trigger denominator disposition, forecast-to-landmark binding, consumer compatibility matrix, and source/authority semantics. **Implementation gate:** no unknown field or lifecycle value silently enters a closed existing schema.

Tests: one writer per identity, exact source-key idempotency, unknown anchor rejection, one-active B1 preservation, typed null, mixed-generation rejection, future-knowledge rejection, correction lineage, stale permission prevention and private-context isolation. **Product:** fixtures and internal preview. **Authority:** all OLI flags false. **Rollback:** remove the optional profile; the existing native payloads remain valid and unchanged.

18.5 P0 – First honest product vertical

Owners: Market OS product owner and existing Prophet/Terminal consumer owners. **Inputs:** C0 plus an already-admitted bounded subject/strategy source path. **Outputs:** one real card/detail/return-change journey with phase, native entry verdict, Plan distinction, geometry, contradiction and freshness; all unsupported forecast heads null. **Dependencies:** no need to pretend broad W2 research is complete.

Tests/proof: exact canonical payload through actual gateway/paywall, API/UI equality, real source/readback, stale and missing cases, causal chart annotation, responsive/a11y/browser receipts. **Authority:** descriptive/contextual only; preserves existing B4's independent authority. **Rollback:** consumer feature flag or optional section removal; no Plan/position mutation or owner data deletion.

18.6 S0 – Lifecycle science v1 registration

Owners: TOI science, Prophet strategy owners and Evaluation/TrialLedger. **Inputs:** accepted W1 method/equivalence inventory, C0, qualified data capability map and current kill registry. **Outputs:** a small admitted first tournament with exact species, population, first forming/trigger rules, landmarks, transition outcomes, baselines, execution/cost policy, search budget, numeric economic margins, sample/power plan and formal look schedule.

Start with one appropriately admitted technical family, such as the W0-prioritized compression/activation question, only after the owner selects its exact definition. The four draft W3 families and 32 constructions do not become registrations by being repeated in this report. Any existing declared trial budget survives; a new family requires a distinct accepted registration. **Tests:** outcome-blind manifest validation and adversarial label fixtures. **Authority:** research only. **Rollback:** retire a defective registration prospectively with a receipt; never silently change its original rules.

18.7 H0 – Historical/PIT evaluation

Owners: existing evaluation/harness owners with independent review. **Inputs:** S0 registration, D0 admitted family, immutable dataset and configuration hashes. **Outputs:** reproduction receipt, denominator ledger, chronological baseline/challenger results, calibrated-head evidence where earned, cost/coverage sensitivity, negative findings and falsifier ruling. **Gate:** exact source availability class; retrospective-only information stays in an explicitly retrospective arm.

Tests: prefix invariance, no future features or matured-label leakage, full-population accounting, same-bar ambiguity handling, splits by real intervals, reproducible seed/config, multiple-testing budget reconciliation.

Product: evidence detail or research lab, not automatic public confidence. **Authority:** historical evidence only.

Rollback: quarantine invalid artifacts and retain original bytes plus correction lineage; no protected-outcome rescue tuning.

18.8 F0 – Prospective forward proof

Owners: existing Radar/episode/QLedger/Trial owners; relevant active Fable repair remains incumbent. **Inputs:** accepted private transport, native event identity, admitted source/model version, registered epoch and validated consumer. **Outputs:** one real untouched event from producer to durable record to grade join, then natural accrual with counts, missingness and integrity receipts.

Tests/live proof: actual R2/runner intake contract, atomic source generation, idempotent replay, prefix preservation, source-present versus source-unavailable empty states, model/config hash at decision, delayed outcomes and exact declared-horizon grading. **Authority:** shadow only. **Rollback:** disable the consumer or new forecast emission while preserving append-only records. A repair creates an honest new epoch where required; it does not relabel prior incomplete transport as prospective success.

18.9 M0 / E0 – Management and accountable re-entry

Owners: Prophet Plan/strategy and B1 owners, TOI/Radar facts, existing Evaluation. **Inputs:** immutable entry law, native position context, admitted current-landmark evidence, registered hold/reset definitions. **Outputs M0:** plan-specific review state and remaining risk, with entry/management factorial comparisons. **Outputs E0:** same-episode landmark versus new-generation law, linked original failure, new anchor/trigger/invalidation and repeated-failure evidence.

Tests: saved-not-entered, entered-but-no-fresh-entry, ordinary volatility versus owner invalidation, expiry, stale source, failed episode preserved, terminal overwrite rejected, duplicate source-key rejection and re-arm suppression. **Product:** Plans and chart/detail. **Authority:** observation/review first; no automatic sell/trim/re-entry. **Rollback:** remove optional assessment, preserve native Plan and episode histories and prior forecast records.

18.10 X0 / P1 / A0 – Incremental evidence, integration and authority

X0 owners: each native evidence owner plus Evaluation. Add theme/breadth, catalyst, regime, options or analog families one at a time under paired, same-information incremental tests. A family that improves prose but not predictions remains descriptive. No broad new fitting is inferred from the completed options retrospective package. [128]

P1 owners: Market OS and existing consumers. Integrate only qualified heads across Candidates, Plans, Terminal, Radar and alerts; prove common reasons, clocks, confidence labels and correction behavior. Use real transport and browser receipts, not screenshots alone. **Rollback:** disable only the new profile/consumer path and keep canonical saves and histories intact.

A0 owners: existing species/evaluation/policy accepting authorities. Decide separately for display, research input, ranking, admission/gating, sizing, alerts and action policies. Require predeclared predictive/economic evidence, calibrated selected-population behavior, source/operational health, feasible costs and rollback. **No blanket OLI promotion.** A rejected family is a completed scientific result, not unfinished engineering to be tuned until it passes.

19. Immediate recovery plan for #7107 / #7094 and projections

19.1 #7107: preserve custody, finish the bounded repair

Refresh the exact branch head, current base, open review and legitimate writer before any edit. Recover the incumbent uncommitted AST validator change and its two tests without overwriting the worktree. The old source-custody/authentication hold is not permission to bypass credentials or create a duplicate branch. Where custody is unresolved, record the blocker and continue independent contract/design work.

Then fix only the outstanding semantics. The generated-ID validator must accept legitimate generated catalog IDs without admitting arbitrary IDs. The hostile duplicate fixture must reach the intended duplicate-membership guard: keep a separate test for two first-class definitions rather than broadening an assertion until it passes. Bind Bollinger bandwidth to the actual numeric primitive and record the remaining integration coverage honestly. Preserve Donchian frozen-occurrence versus latest-break resetting as a partial unless a separately approved compatibility contract proves equivalence.

Synchronize the method, class, tier and report counts; proposed 24/3/5 must match the final actual classifications before publication. Preserve the earlier ConnorsRSI/RVOL accepted review and existing trial budgets. Finally establish current-base compatibility, focused/full relevant tests, hosted exact-head checks, independent review and explicit W1 owner acceptance. Only that accepted handoff selects what may proceed to W3. [I05, I06]

19.2 #7094: do not confuse records acceptance with data admission

Refresh the incumbent exact head and current main. Close remaining release/custody/check gates for the already-reviewed archaeology record through normal protected review. Its output remains PARTIAL/HOLD. Reconcile the Agent OS state through the existing carrier's planned record changes rather than creating an overlapping "repair" elsewhere.

Release the bounded W2 repair only after the rightful owner accepts the preceding record. Make actual exchange close load-bearing; prove early-close final-bar semantics; reject unsupported hourly substitutions; qualify a common Daily/4H basis and correction family; publish historical eligibility, coverage and unknowns. Use deterministic aggregation fixtures before any outcome study. A passing regular-session sample cannot waive early closes, corporate actions, missing history or broader universe qualification. Return the admitted family and explicit exclusions to W3; do not declare all timeframes globally solved. [I07, I08]

19.3 Projection corrections

Agent OS should show W1 and W2-0 as existing draft/held carriers with their exact acceptance debt. Linear P-MAS-54 should link those real carriers and their owners rather than remain an empty proxy for nonexistence. Temporal W0 should reflect #6790's merge without implying later research completion. Top Anatomy should reflect restoration complete plus deliberate natural wait. #784's active incumbent receipt should supersede old receiverless text. These are future bounded records actions; none was performed in this research run. [I03, I10, I11, I14, I15, I24, I25, I26]

19.4 Recovery acceptance packet

The release packet needs an immutable source pin, reviewed head, changed-path inventory, preserved-dirty-work receipt where relevant, tests and hosted check IDs, review ruling, owner acceptance, actual capability state, remaining exclusions, and exact next action. "PR exists," "tests passed locally," "merged," "running," "data qualified," and "validated" must remain distinct labels.

20. Kill and defer list

Do not build a universal OLI score or a single stock lifecycle enum. They collapse incompatible phases, policies, positions and evidence. **Do not create new episode, event, market-data, options, theme, trial, memory or lifecycle stores.** Demonstrate a specific existing-owner gap before proposing a minimal schema extension. **Do not re-create W1/W2, the B1 anchor allowlist, #784's reclaim execution, accepted options measurement, or Top restoration.** [I04, I05, I06, I07, I08, I09, I14, I15, I16, I17, I18, I22, I24, I28, I30]

Defer hidden-state and sequence models until a transparent model has a specific unresolved failure. Defer adaptive per-stock horizons until Temporal Grain's causal usefulness path is admitted. Defer options-based direction, dealer-inventory assertions, exact-option strategy returns and broad fusion until their own source/economic gates pass. Defer broad international deployment of forecast claims until market-specific source and execution contracts qualify.

Kill hindsight-derived phase labels, retrospectively selected "good resets," target/stop order invented from coarse bars, successful-confirmer-only comparisons, current-universe historical membership, and retrospective data relabeled prospective. Defer auto-sell, auto-trim, auto-re-entry and capital allocation: management intelligence is not action authority. Do not normalize a failed first episode into a profitable long thesis.

Do not build a new evaluation framework to make the research easier. Reuse the existing trial, claim, species and arena owners. Do not run massive construction searches while the data substrate is held. Compute cannot fix missing knowledge-time provenance, wrong session boundaries or an unrecorded source population.

21. Open questions, falsifiers and decision gates

21.1 Questions that require owner decisions

The proposed semantic home needs explicit adoption by the existing Alpha/TOI/Market OS owners. The permitted OEV extension point and K5 dependency boundary must be confirmed. The native owner for first-forming denominator persistence and forecast-to-landmark linkage must be selected after inspecting current interfaces. Prophet B1 must decide whether any new anchor species or simultaneous economic episode capability is actually necessary; current rules remain binding until then.

The exact first species definition, economic minimum improvement, harmful-tail margin, useful coverage floor, sample/power plan, historical source eras and formal look schedule must be frozen before outcomes. These are intentionally not guessed. An implementation lead can begin recovery and contract work immediately without treating an incomplete numerical registration as permission to fit.

21.2 Explicit falsifiers

Hypothesis	Evidence that would reject or narrow it	Required response
Lifecycle modeling adds information	Same-information static/rule/hazard baselines match the richer model on untouched data	Keep the simpler model; retain only proven explanatory/product value.
Current-landmark updates improve decisions	Forecast metrics improve but feasible policy value does not, or costs remove the advantage	Do not promote to action/rank; retain research-only forecasts if useful.
"Consumed opportunity" is the entry failure mechanism	Progress/extension adds no incremental value after controls, or winning tails are disproportionately removed	Reject a blanket extended veto; preserve current-entry uncertainty.
Hidden fragility explains holding failures	Fixed-entry comparisons show no reproducible improvement from the preregistered context	Retain negative result; no subgroup/horizon rescue in that epoch.
Re-entry refreshes opportunity	Advantage depends on dropping earlier failures or changing old anchors/horizons	Reject the re-entry method and correct the evaluation lineage.
Added breadth/catalyst/options helps	Gains disappear under price/group/coverage controls or source-era holdouts	Keep descriptive context only, or retire the feature family.
Adaptive clocks improve usefulness	Advantage vanishes under instrument-disjoint tests or equal-memory controls	Retain fixed qualified clocks; do not choose best timeframe per ticker.
Product integration helps users	Users still confuse entry/hold/Plan state, source freshness or probability meaning	Revise or remove the misleading surface before expanding.

21.3 Inconclusive is not success or failure by default

An immature OOT window, an unqualified data family, a sparse competing event, a broken transport path or a failed source join can prevent a scientific verdict. Record the exact reason. A qualified negative result may falsify a model; an unavailable observation cannot. Conversely, engineering success does not justify a provisional claim of edge while evidence accrues.

21.4 Completion criteria for this commission versus execution

This report resolves the architecture recommendation, ownership boundaries, ontology proposal, identity law, research design, product composition, recovery sequence and handoff. It does **not** resolve the market's empirical answer, perform W1/W2 acceptance, start a prospective epoch, recertify all live production data, or promote a trading policy. Those are deliberately explicit future evidence gates rather than hidden promises inside the research conclusion.

22. Next-execution handoff

Mandate

Execute **recovery and a bounded, source-honest opportunity-context vertical** through existing owners. Do not start a new OLI control plane. Treat this report as a research/design recommendation requiring native owner acceptance, not as an override of current protected procedure or an authorization to consume protected outcomes.

First bounded work package

Objective: reconcile TOI's real carriers and obtain a reviewed contract for one observational cross-owner view.

Logical principal: existing Sol/Alpha/TOI authority coordinating Market OS and Prophet; use Fable for genuinely cross-owner execution where already assigned or lawfully placed. Preserve #784's live Fable principal. Do not create another assignment simply because a research package now exists.

Start inputs: current protected Mastermind procedure; exact current heads of Macro/Terminal; #7107/#7094/#7069 and their reviews; Alpha K3-E and Market OS ownership decisions; current B1 code; actual #784 receipts; relevant current kill registry entries. The source pins in this report are reproducibility anchors, not permission to skip a fresh collision census.

Required outputs: (1) owner-approved reconciliation with real statuses and next actions; (2) exact W1/W2 custody/repair disposition; (3) schema/identity compatibility review for a governed OEV opportunity-context profile; (4) pre-trigger denominator and landmark-record owner ruling; (5) one bounded product fixture and planned real-source proof path; (6) a separately frozen first-study registration plan with unresolved numeric gates named, not fabricated.

Forbidden changes: new registry/store/scanner/score; cancellation or absorption of #784; modification of Top's OOT ruler; broad options fitting or direction promotion; opening W3 outcomes before W1/W2 gates; automatic Plan/position writes; silent schema widening; credential workarounds; replacing accepted UI/source; or use of Vercel.

Handoff acceptance and continuity

A successful handoff returns the exact carrier/head, accepted owner decision, tests/reviews, capability/evidence class, live proof if applicable, authority ceiling, blocker, and next action. Partial work must identify its incumbent writer and custody; a successor must not reconstruct or overwrite it. GitHub remains implementation/evidence truth, Agent OS organizational continuity, Linear the portfolio projection, and the Executive runtime a separate installed/running-state observation.

The immediate next action is **R0 plus custody-safe D0 planning and C0 contract adjudication**. The first customer-visible goal is PO's truthful lifecycle context—not a new numeric probability. The first scientific goal is a registered, qualified, same-information landmark test—not a larger construction catalog. This sequence recovers the existing program while preserving the evidence required for it eventually to become genuinely better.

Source register

All sources below were consulted or retrieved during this commission on October 3, 2026. An access description is not a claim that a full paper, entire code module, production runtime, or purchased product was audited. Historical receipts retain their stated dates. Source pointers on private GitHub/Linear require the reader's normal access.

Internal sources and observations

[C00] Chairman commission: Mastermind Opportunity Lifecycle Intelligence — Pro Deep Research Commission

Uploaded text, fully supplied. 1,126 source lines; SHA-256 366d57bceaf2fded1780d88862e2d424a7251817f3da0dd1d5ec4812b1e45944

[I01] Remote source references: Mastermind master; Macro main; Terminal master

Native connector read. Mastermind 50a7a5d31854c1161335e2b6ffdb04eb268fe6d; Macro 6f5e78e94e8808582a650cdfa0fc3357040a179c; Terminal 41b8af2da46614cedd2a485214e53003d4f030fc. Read sequentially, not an atomic three-repository snapshot. GitHub API git/ref reads.

[I02] Mastermind Executive V2 executive_state observation

Native executive_state read. 2026-10-03T08:56:52Z; installed Mastermind b3627c580dd37ac1c167f59ac4b7555a4330edef; installed Macro 88804ed7079700c598bb8e04aa64307d1335402d. Seven jobs: four completed, three failed; no running jobs in this returned view. This is not a census of every external Fable session.

[I03] TOI Agent OS workstream

[Source](#) · Native connector read. Current-main organizational record; W1/W2-0 undispatched text conflicts with live PRs.

[I04] TOI W0 architecture freeze

[Source](#) · Native connector read. Architecture and first 260 lines inspected; historical W0 law, not a newly validated model.

[I05] TOI W1 incumbent research carrier

[Source](#) · Native connector read. Open/draft; head 15e4dfb0c9ddc788127d0954fd7d92414eefdcfb. Research counts and prior test results are reported receipts, not re-executed here.

[I06] TOI W1 independent CHANGES_REQUESTED review, September 28, 2026

[Source](#) · Native connector read. Native list_pull_request_reviews read: generated-ID validator, duplicate fixture, numeric Bollinger binding, frozen-occurrence Donchian mismatch. Existing uncommitted AST repair has custody; preserve prior accepted ConnorsRSI/RVOL review.

[I07] TOI W2-0 incumbent data/clock archaeology carrier

[Source](#) · Native connector read. Open/draft; head 5bb1bc68c99146fab040aade04bbf1903c51e5b7; 15/15 regular-session parity passes and 5/5 early-close failures are reported evidence.

[I08] TOI W2-0 exact-head semantic reviews, September 13, 2026

[Source](#) · Native connector read. Native reviews: semantic PASS / RELEASE_BLOCKED. A records merge would accept PARTIAL/HOLD archaeology, not admit the data substrate or W3.

[I09] TOI Elliott E0 source finding / admission proposal

[Source](#) · Native connector read. Open/draft; head 3aa54c741aa6218379ca9498069219f84d876d2b; SPEC_ONLY / NOT_ADMITTED / HOLD_FOR_TOI_W1_AND_W2_0.

[I10] Linear P-MAS-54: Technical Opportunity Intelligence

Native get_project and list_issues, including archived issues. Project In Progress; child result empty, hasNextPage=false. Last recorded projection September 3.

[I11] Linear P-MAS-51: Top Anatomy

Native get_project and list_issues, including archived issues. Project In Progress; child result empty, hasNextPage=false. Empty children do not imply absent work.

[I12] Linear MAS-25: Sol post-merge acceptance — Live Entry Radar W6

[Source](#) · Native connector read. In Review; W6 implementation on main through #5834/#5845, final owner acceptance still owed in the retrieved record. Distinct from MAS-24.

[I13] Terminal #598: Tactical Intelligence parent

[Source](#) · Native connector read. Open; latest metadata update October 3. Original kickoff body is historical; active child #784 must be considered.

[I14] Terminal #784: Intraday Dislocation + Reclaim incumbent

Source · Native connector read. Active Fable pickup and START receipts supersede older no-receiver handoff text. Do not duplicate or absorb the execution.

[I15] Terminal #784 pickup, START, and source audit

Source · Native connector read. Pickup comment 5967226402 and START comment 5967226570, October 3. W4 R2/W5 absent local-directory intake mismatch and catalyst defects are the active incumbent audit's findings, not rerun here.

[I16] K3-E Opportunity Evidence Vector contract decision

Source · Native connector read. Typed, deterministic view/join; no new store. Existing lifecycle labels were not admitted into v1. Generic owner_pit_reference is retrospective-only.

[I17] Accepted K3-E OEV contract and executable semantics

Source · Native connector read. Merged August 27, a1bdf2a2ad051cc63a9a5070da11057fe9ddb6fc. Old HOLD prose does not override merged metadata; merge is not proof that all consumers are live.

[I18] No-new-workstream productization ownership decision

Source · Native connector read. Alpha Intelligence owns K1/K3/K5 semantics; Market OS owns security_state product execution. OpportunityCase and security_state are distinct.

[I19] Market OS workstream and historical live acceptance

Source · Native connector read. Current record of A1A/A1B and AAPL B1A acceptance. Broad namespace/identity expansion remains gated in the record; no fresh browser recertification this run.

[I20] Prophet US effective R6 master plan

Source · Native connector read. Effective September 23; proposal and execution plan, not blanket release. Sections on ownership, origin/current landmark, entry, holding, measurement and product inspected.

[I21] Prophet R6 B02 episode-registry census, September 23

Source · Native connector read. Historical census at aba2c115...; its open anchor-vocabulary defect is superseded by current code I22. Historical grader coupling findings are not a new live proof.

[I22] Current canonical candidate episode core

Source · Native connector read. Direct code read: closed ANCHOR_KINDS={turn_watch_reset_low}; immutable ordinary source keys; one active episode per security/identity epoch; exact Radar attachment; causal opened_at. Blob 525a5bb7cea6a5f84fe23999e53a82c12a08b5c1.

[I23] Prophet Phase 22 fast-cycle × hidden-fragility prospective preregistration

Source · Native connector read. Records-only frozen specification. Start, transport, rates known-at and TrialLedger gates remain explicit. Existing floors are preserved, not generalized to all OLI studies.

[I24] Top Anatomy workstream

Source · Native connector read. R0 restoration complete; frozen OOT naturally accruing, earliest eligibility roughly July 2027. Historical receipt, no new outcome peek.

[I25] Temporal Grain workstream

Source · Native connector read. Current W0 awaiting-CI projection is stale relative to merged #6790. Mechanical W1A and usefulness W1B are distinct gates.

[I26] Temporal Grain W0 architecture merge

Source · Native connector read. Merged September 3 as db5d20c45db123a2e133d9c1a28387ec9f23a545; SPEC_ONLY / PRODUCTION_INERT.

[I27] GMI Theme Graph workstream

Source · Native connector read. Graph substrate and active theme verticals exist. D2C/D2D/D2E/W3B/W3C have explicit completion/acceptance dependencies; current universal theme-state qualification not inferred.

[I28] Options Alpha recovery workstream

Source · Native connector read. Includes Oct 2 historical research closeout: 60 cells, three within-family BH rejections, all PIT/OOS/ promotion boundaries retained. Natural measured Sep 17 evidence is do-not-redo; publication proof remains separate.

[I29] Prophet R6 earnings/D5 readiness census, September 23

Source · Native connector read. Historical narrow AAPL dossier/adaptor proof with injected owner-read seams; three allowlisted lanes; consensus absence and dual knowledge clocks. Not whole-universe live proof.

[I30] Intelligence Evaluation OS architecture

Source · Native connector read. Historical architecture identifies existing QLedger, Arena, Species, Trial and kill owners. Historical inventory counts are not presented as current counts.

[I31] Protected Mastermind Skillpack INDEX / cold-start / active-execution procedure

Source · Native connector read. Current protected source re-pinned and procedure read. Source acceptance, runtime installation, production proof and scientific promotion remain separate.

[I32] Existing Opportunity Evidence composer

Source · Native connector read. Existence and contract role located by native source search; contract decision and merged carrier are the primary authorities, not an independent full-module test.

Academic and methodological research

[X01] Lo, Mamaysky & Wang (2000), Foundations of Technical Analysis

Source · Primary paper/publisher or author source. Official working-paper abstract and journal metadata; no new replication.

[X02] Jegadeesh & Titman (1993), Returns to Buying Winners and Selling Losers

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X03] George & Hwang (2004), The 52-Week High and Momentum Investing

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X04] Lee & Swaminathan (2000), Price Momentum and Trading Volume

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X05] Hong, Lim & Stein (2000), Bad News Travels Slowly

Source · Primary paper/publisher or author source. Official paper abstract; NBER working paper issued 1998.

[X06] Moskowitz & Grinblatt (1999), Do Industries Explain Momentum?

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X07] Daniel & Moskowitz (2016), Momentum Crashes

Source · Primary paper/publisher or author source. Official working-paper abstract; working paper issued 2014.

[X08] Ng, Rusticus & Verdi (2008), Implications of Transaction Costs for the Post-Earnings Announcement Drift

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X09] Pan & Poteshman (2006), The Information in Option Volume for Future Stock Prices

Source · Primary paper/publisher or author source. Publisher abstract; specialized buyer-initiated opening-volume dataset.

[X10] Zaremba, Szyszka, Karathanasopoulos & Mikutowski (2021), Herding for Profits

Source · Primary paper/publisher or author source. Publisher abstract and author-deposited paper metadata; no pooled effect-size analysis.

[X11] van Houwelingen (2007), Dynamic Prediction by Landmarking in Event History Analysis

Source · Primary paper/publisher or author source. Publisher abstract and method description.

[X12] Putter, Fiocco & Geskus (2007), Tutorial in Biostatistics: Competing Risks and Multi-State Models

Source · Primary paper/publisher or author source. Publisher/PubMed abstract; full publisher retrieval unavailable.

[X13] Fine & Gray (1999), A Proportional Hazards Model for the Subdistribution of a Competing Risk

Source · Primary paper/publisher or author source. Publisher metadata and methodological summary.

[X14] Gneiting & Raftery (2007), Strictly Proper Scoring Rules, Prediction, and Estimation

Source · Primary paper/publisher or author source. Author-hosted full PDF text; first-page screenshot and scoring-rule sections inspected.

[X15] Gerds & Schumacher (2006), Consistent Estimation of the Expected Brier Score in General Survival Models with Right-Censored Event Times

Source · Primary paper/publisher or author source. Publisher/PubMed abstract.

[X16] White (2000), A Reality Check for Data Snooping

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X17] Hansen (2005), A Test for Superior Predictive Ability

Source · Primary paper/publisher or author source. Publisher and author research-page summaries.

[X18] Sullivan, Timmermann & White (1999), Data-Snooping, Technical Trading Rule Performance, and the Bootstrap

Source · Primary paper/publisher or author source. Publisher abstract/metadata.

[X19] Harvey, Liu & Zhu (2016), ... and the Cross-Section of Expected Returns

[Source](#) · Primary paper/publisher or author source. Official working-paper abstract; working paper issued 2014.

[X20] Bailey, Borwein, López de Prado & Zhu (2017), The Probability of Backtest Overfitting

[Source](#) · Primary paper/publisher or author source. Author/university repository; DOI 10.21314/jcf.2016.322.

[X21] Yu (2010), Hidden Semi-Markov Models

[Source](#) · Primary paper/publisher or author source. Publisher methodological review abstract; duration-model precedent, not finance evidence.

[X22] Adams & MacKay (2007), Bayesian Online Changepoint Detection

[Source](#) · Primary paper/publisher or author source. Author preprint and author-lab summary.

[X23] Lim, Arık, Loeff & Pfister (2021), Temporal Fusion Transformers

[Source](#) · Primary paper/publisher or author source. Publisher full HTML including inputs, model components and interpretation sections.

[X24] Gu, Kelly & Xiu (2020), Empirical Asset Pricing via Machine Learning

[Source](#) · Primary paper/publisher or author source. Official paper abstract; NBER working paper issued 2018, revised 2019.

[X25] Kaelbling, Littman & Cassandra (1998), Planning and Acting in Partially Observable Stochastic Domains

[Source](#) · Primary paper/publisher or author source. Publisher abstract; theoretical decision-framework precedent.

[X26] Leung & Li (2015), Optimal Mean Reversion Trading with Transaction Costs and Stop-Loss Exit

[Source](#) · Primary paper/publisher or author source. Publisher abstract and author preprint. Ornstein–Uhlenbeck spread assumptions, not generic equity law.

[X27] Gibbs & Candès (2021), Adaptive Conformal Inference Under Distribution Shift

[Source](#) · Primary paper/publisher or author source. Official NeurIPS abstract; long-run coverage is not per-state conditional calibration.

[X28] Geifman & El-Yaniv (2019), SelectiveNet

[Source](#) · Primary paper/publisher or author source. Official ICML/PMLR abstract; risk–coverage framing, not trading evidence.

[X29] Putter & van Houwelingen (2022), Landmarking 2.0

[Source](#) · Primary paper/publisher or author source. Publisher/PMC full HTML available; dynamic-covariate and landmarking versus joint-model framing.

[X30] Derman (2002), The Perception of Time, Risk and Return During Periods of Speculation

[Source](#) · Primary paper/publisher or author source. Author preprint; conjectural intrinsic-time model, not validated adaptive horizon rule.

Professional product documentation

[P01] TradingView: strategy alerts

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P02] TradingView: repainting and custom-script alerts

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P03] TradingView: historical order time versus real-time alert time

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P04] TrendSpider: multi-factor alerts

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P05] TrendSpider: Strategy Tester setup and constraints

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P06] TrendSpider: shared visual scripts across scanner, backtest and alerts

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P07] MarketSurge: developer-published app description

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P08] Trade Ideas: Holly risk modes

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P09] Koyfin: 2025 product updates, published 2026

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P10] Koyfin: Watchlist News

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P11] TC2000: shared condition editor

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P12] TC2000: price alerts and restart parameters

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P13] Bloomberg: Research Management Solutions

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P14] Bloomberg: Portfolio and Risk Analytics

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P15] LSEG Workspace: equity research and Watchlist Pulse

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P16] LSEG: StarMine SmartEstimates

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P17] FactSet: research-management integration

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P18] FactSet: Intelligent Platform / IRN 2.0 announcement

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.

[P19] TrendSpider: confirmation candle and trigger semantics

[Source](#) · Official current documentation or vendor-published description. Documentation review only; no authenticated product testing or independent validation of performance claims.